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Date: Certified by:	Space reserved for supplier's certified stamp	Document Acceptance code: 1 ☐ Accepted 2 ☐ Accepted incl comments 3 ☐ Not accepted revise and resubmit 4 ☐ For information Date: _____ _____Signature
Equipment Description BQ-S Mobile Battery Container 10ft		
Supplier Doc no / Area / System		
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Document title.

User Manual BQ-S

Installation:	Total no sheets: 92	PSW P&A document number 238833-5341-E-MA-0001

POWER & AUTOMATION
BQ-S BATTERY CONTAINER 10ft



USER MANUAL

Customer: PON Energy Rental (5341)

Revision 03

18.01.2024

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1 Introduction

1.1 Description

The purpose of this document is to describe how to safely use and operate the BQ-S battery container.

PSW Power & Automation has developed BQ – "Battery Intelligence", which is a dynamic energy storage system for hybridization and energy optimization for power producers and power grids.

The battery solution is used to optimize power consumption by providing power boosts and energy storage leading to energy cost optimization.

- BQ - Energy Optimization, using load smoothing, improves the operation and utilization of the power sources. If an energy producer has a shut down, the battery system can handle the loads until a new power source is online. In addition, if there is excess energy the power producer can charge the battery as needed.

- BQ – Power Boost prevents energy producer overload by using the battery to handle peaks. The energy producer provides power within its capabilities, while the battery provides load smoothing.

	***SAFETY INFORMATION!*** *Before connection and operation, it is important that the personnel involved read the user manual carefully. The operator must have satisfactory training before starting the plant.*

SAFETY	
General	*Please read the documentation accompanying the project so that you are familiar with the safety signs and instructions before using the equipment.* *This product is designed and tested in accordance with international standards. The equipment must be used for the intended use.* *Warning – The system must only be maintained by qualified personnel. To reduce the risk of electric shock, do not perform any service other than that specified in the manuals unless you are qualified to do so.* *If you open the container / cargo doors during operating mode, there will be an automatic shutdown sequence after a set time (delay in software application).*
In case of fire	*DO NOT ENTER THE BATTERY ROOM WHEN THE FIRE DETECTION IS TRIGGERED OR SMOKE IS VISIBLE FROM THE VENTILATION OR ROOM* *In the event of smoke or fire from the battery compartment. Leave the battery room immediately*
NOTICE	*Notify the fire brigade immediately if smoke is detected from the battery compartment.* *Hand over: Material Safety Data Sheet for the fire service personnel*
Other information	*Extinguishing equipment: Water, CO2, foam, powder, sand clay* *Special fire-fighting procedures: Self-contained breathing apparatus.* *The cell can be vented if exposed to excessive heat.* *Hazardous combustion products: Carbon monoxide, carbon dioxide, lithium oxide vapour*

1.2 Health & Safety

For safe installation, commissioning, and use of the BQ, it is important that HSE requirements are followed.

In addition, HSE rules from the owner of the BQ container must be followed.

The product is used in combination with a permanent energy source (battery). Even if the equipment is switched off, a dangerous electrical voltage may occur at the input and/or output terminals.

Operation of the plant

- No loose parts are allowed on the floor. No storage of components is allowed in the room.
- Wear PPE when working on electrical equipment.
- Always keep the cargo doors closed.
- Pay attention to automatically regulated fans.
- Only qualified electrical personnel can perform service and maintenance on the system.
- All covers and cabinet-doors / lids / hatches must be fitted and always closed.
- Follow this manual for safe commissioning, operation and maintenance.

1.3 Symbols

In this manual, these symbols are used to indicate important INFORMATION or DANGER.
Read the instructions carefully.

	ELECTRICAL HAZARD Great danger if used incorrectly
	WARNING Important WARNING for safe use
	INFORMATION Important information under this symbol

1.4 Abbreviations

This document contains abbreviations. The following may be used in the text.

Abbreviations	
DC	Direct Current
EM	Emergency stop
GA	General Arrangement
HMI	Human Machine Interface
HSE	Health, Security and Environment
ISO	International Organization Standardization
kVA	Kilo Volt Ampere
MRB	Master Record Book
MVA	Mega Volt Ampere
KWH	Kilo Watt Hours
PLC	Programmable Logic Controller
PE	Protective Earth
SJA	Safe Job analysis
SPC	Shore Power Container
TN-C	Terra Neutral Combined Network-grid
VFD	Variable-frequency drive
VPN	Virtual Private Network
UPS	Uninterruptible Power Supply
BMS	Battery Management System
SoC	State of Charge in %
RBMS	Rack Battery Management System
KPMC	Kore Power Management Controller
PMS	Power Management System
ESU	Energy Storage Unit
IGBT	Insulated Gate Bipolar Transistor
OCV	Open Circuit Voltage

1.5 General



It is not permitted to leave the container doors open during operation. It is also forbidden to remain inside the container during operation. To avoid health risks, it is recommended to only stay outside the container with the doors closed.
There is a door contact switch inside the container, which shuts down the system for safety reasons when open. It is forbidden to manipulate this sensor.

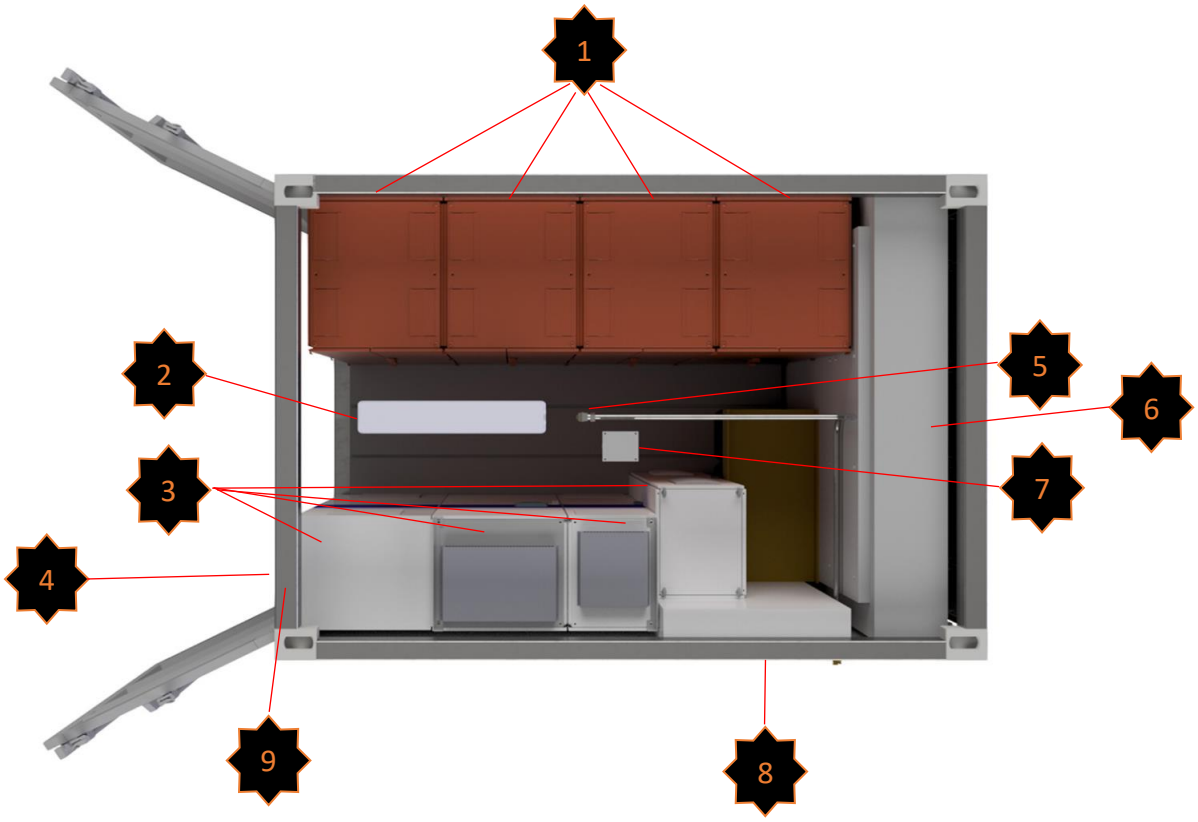


Inside the container, all active parts are protected with IP2X contact protection, but it is still recommended to always think of safety first and to comply with all measures to maintain health. Make sure to never work when the system is running or under voltage.

To ensure trouble-free operation of the BQ, it is essential that the operating personnel know the system well. Instructions and manuals must be read and understood.

	INFORMATION Only qualified and instructed personnel can start up and operate the BQ.

Main components in the system:



No.	Component / Description	Comment
1	Battery Cabinet Racks	String 1-4
2	Container Illumination	With motion detection
3	Cabinets	Control, VFD, DC
4	User Operating Interface	HMI, Display
5	Fire Water Outlet	
6	Reefer unit	Heating & Cooling
7	Explosion hatch	In the roof from Container
8	Connection Interface	AC Input / Output
9	Door Switch [SF1]	Top of the cabinet

2 Installation

2.1 General

	WARNING Always check the routing of the cables for sharp edges, reduced bend radius or lack of protection. Correct any errors before starting the unit.

	WARNING It is important that the doors are closed. This is to avoid water penetration into the container and also prevents entry of pests.

	ELECTRICAL HAZARD The PE connection cable must be correctly dimensioned and have good, corrosion-free, contact on the terminals. If in doubt, perform a grounding measurement.

	ELECTRICAL HAZARD The unit can deliver high power on the outputs. It can cause serious injury if faults occur. Make sure that the entire system, including external equipment, is securely connected and covered before power is delivered from the consumer.

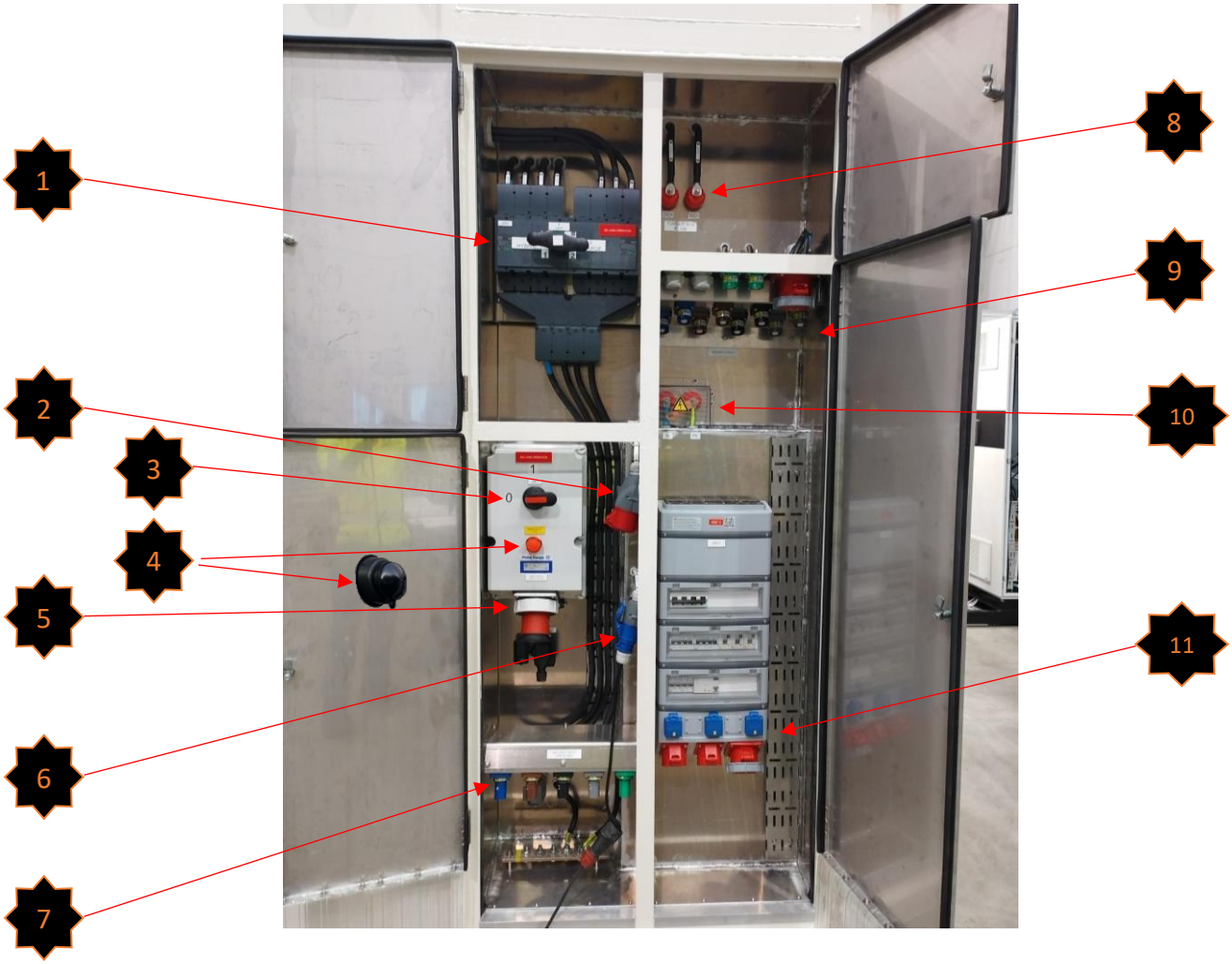
	WARNING Do not have the 125A CEE Switch (QB1) “ON” and the 325A Powerlock Selector Switch (QB2) in “Charge” position at the same time as this will generate a short circuit!

2.2 Unit Datasheet

Unit Datasheet	
Storage Supply Connection – CEE	208-240VAC, 50Hz, 16A 1P+N+PE
Charge Connection – CEE	380-440VAC, 50-60Hz, 125A, IT/TN 3P+N+PE / CEE
Charge Connection – Power Locks (Input)	380-440VAC, 50-60Hz, 325A, IT/TN 3P+N+PE / Power Locks
Charge/Discharge Connection – Power Locks	660-690V, 50-60Hz, IT/TN, 264A - 276A 440-480V, 50-60Hz, IT/TN, 379A - 413A 380-440V, 50-60Hz, IT/TN, 413A - 479A 208-240V, 50-60Hz, IT/TN, 385A - 444A 3P+N+PE / Power Locks (current rate based on continues operation)
Discharge Connection - CCB	400V, 50Hz, 16A, 32A, 63A, 125A, TN
Extra Battery Connection - BUSBAR	800-1100VDC, 500A
Installed energy	442kWh
Available energy	350kWh
Nominal apparent power	160kVA (208-240V), 315kVA (380-480V, 660-690V)
Max apparent power <45 min	200kVA (208-240V), 400kVA (380-480V, 660-690V)
Overload	140% (<1min)/160% (<2 sec)
Nominal round-trip efficiency (IEC 62933-2-1)	>82%
IP degree	IP56
Ambient conditions	-20°C to +40°C
Cooling/heating	Air Cooled (Air/Air)
Fire extinguishing	Storz Coupling, 1 ½”
Detection	Smoke detection
Housing type	ISO 10ft HC container
Dimensions L x W x H	3163- 2438 – 2896 mm
Corrosion level	C3-C5
Noise (low-high)	1m distance 63-79 dB
Weight	8.500kg

For more information and details also check the complete datasheet.

2.3 Overview Connections



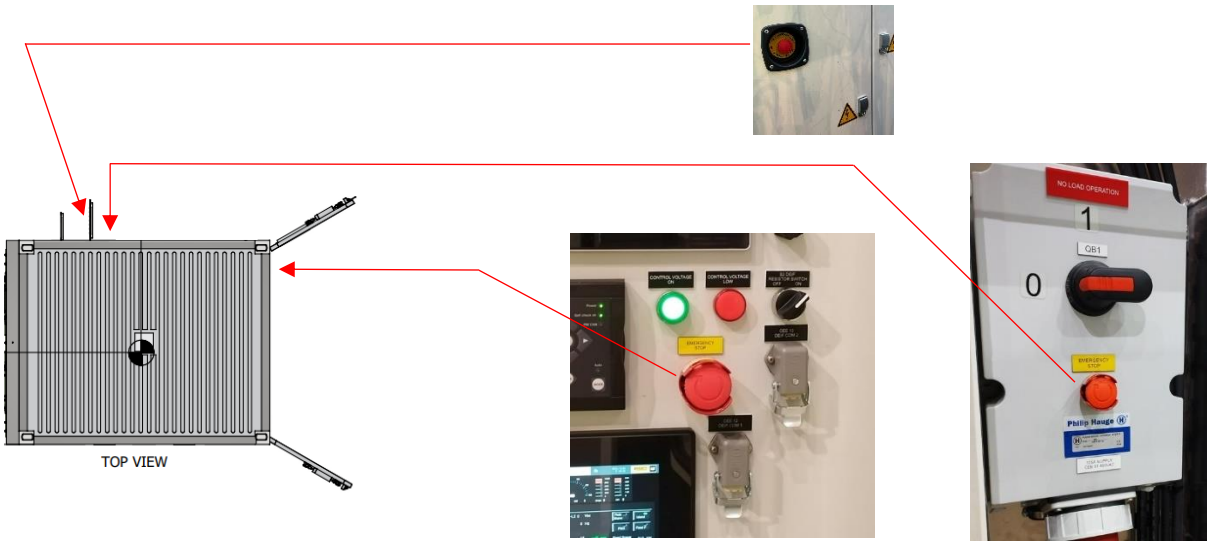
1	QB2 – Powerlock Selector Switch	“Hybrid” – “Off” – “Charge” (No Load Operating)
2	CEE03 – 32A Reefer Supply	400VAC, 50Hz, 32A, 3P+N+PE
3	QB1 - CEE Switch	400VAC for CEE 125A Input plug
4	EM2+EM3 - Emergency Stop Buttons	EM-Stop Buttons Outside
5	CEE01 - 125A Input Connection	400VAC, 50Hz, 125A, 3P+N+PE
6	CEE02 – 16A Storage Supply Connection	208-240VAC, 50-60Hz, 16A, 1P+N+PE
7	Power Locks - AC Connection (1x Set)	380-440VAC, 50-60Hz, 400A, 3P+(N)+PE
8	DC – Connection of additional battery capacity	750-1100VDC, 315A
9	Power Locks – AC Output Connection (2x Set)	660-690V, 50-60Hz, IT/TN 440-480V, 50-60Hz, IT/TN 380-440V, 50-60Hz, IT/TN 208-240V, 50-60Hz, IT/TN 2x 3P+(N)+PE
10	Bridge/Cupper Busbar – Connection N+PE	Network IT/TN
11	Distribution Box - DB1 with AC Output Sockets	Several, for details check “SLD”

3 Handling & Control the System

3.1 Overview

3.1.1 Emergency Stop

There are 3x emergency stop switches outside at the container.
1x Front of container (cargo door hatch)
1x AC interface outside
1x AC Interface behind the door in the CEE 125A Socket



3.1.2 Smoke Detector



There is a sensor for detecting smoke inside the container (ceiling in the middle of container). When activated, this signal is transmitted to the controller via a digital signal which causes the system to shut down. The smoke detector must be checked regularly with a suitable test spray during routine maintenance (6 months) (ref. chapter 8: Maintenance). After an alarm has been triggered, the internal test chamber of the detector must be completely free of particles. For reset after alarm, press the reset button on the HMI display.

The smoke detector is for safety, always ensure that it is not tampered with or covered.

3.1.3 Door Switch (Cargo Door) [SF1]

For security there is a door switch inside the container. It is not allowed to open the cargo doors and enter the container during operating mode. If the door is opened there will be an automatic shutdown of the BQ unit after 30 seconds. Under certain situations trained personnel may need to open the doors. In these cases, on the “customer settings menu” (HMI BQ) there is an “override door switch button” option and also the time to reset the function. In that special case, when you open the cargo doors and want to enter the battery container during operation mode, make sure you use sufficient protective equipment for this environment (ear protection, special protective clothing, isolation / protective helmet with visor, ...). It’s not allowed to tamper with this door switch!

3.1.4 Fire Water Inlet Connection

Outside the container there is a connection point for the fire water inlet.
See the picture below.



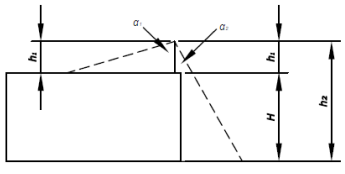
However, before activating the fire water line or increasing the water pressure, confirm the risk. If activated, the batteries, cabinets and interior are sprinkled with water from the sprinkler system on the ceiling. This is primarily used to extinguish possible fires in cables, transformers, etc. but not to extinguish a lithium battery fire.





3.1.5 Lightning Protection Rod

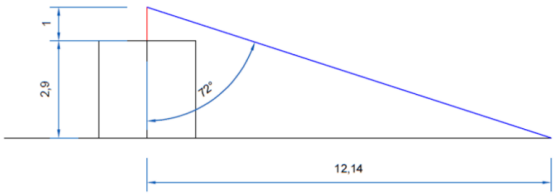
A lightning protection rod is located in the rear of the container. The protection of the simple rod 1m height (ref.110035) is to level III, according to IEC 62305-3, the angle gives protection to a radius of 12,14m.

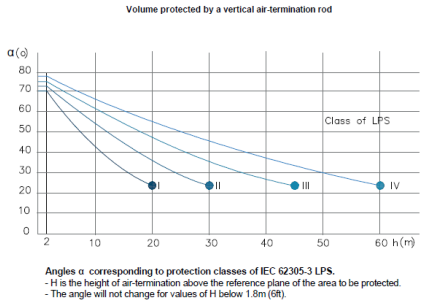


Key

h_i physical height of an air-termination rod

NOTA The protection angle α_1 corresponds to the air-termination height h_1, being the height above the roof surface to be protected; the protection angle α_2 corresponds to the height $h_2 = h_1 + H$ the ground being the reference plane; α_1 is related to h_1 and α_2 is related to h_2.

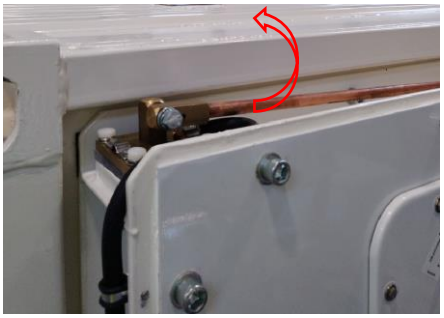








Always make sure that the copper rod is "folded in" and secured for transport.
On arrival at a new location, if outdoors, deploy the lightning protection before putting the unit into operation. This is done by assembling and adjusting the lightning protection rod and establishing a suitable earth connection (lightning arrester).







3.1.6 Outside 3-Color Beacon [EA1]

As an option, a signal tower (EA1, 3-colored LED light) outside the container could be used for indication of faults and warnings.



red	–	Fault / alarm
green	–	No fault or warnings.
yellow /orange	–	Warning

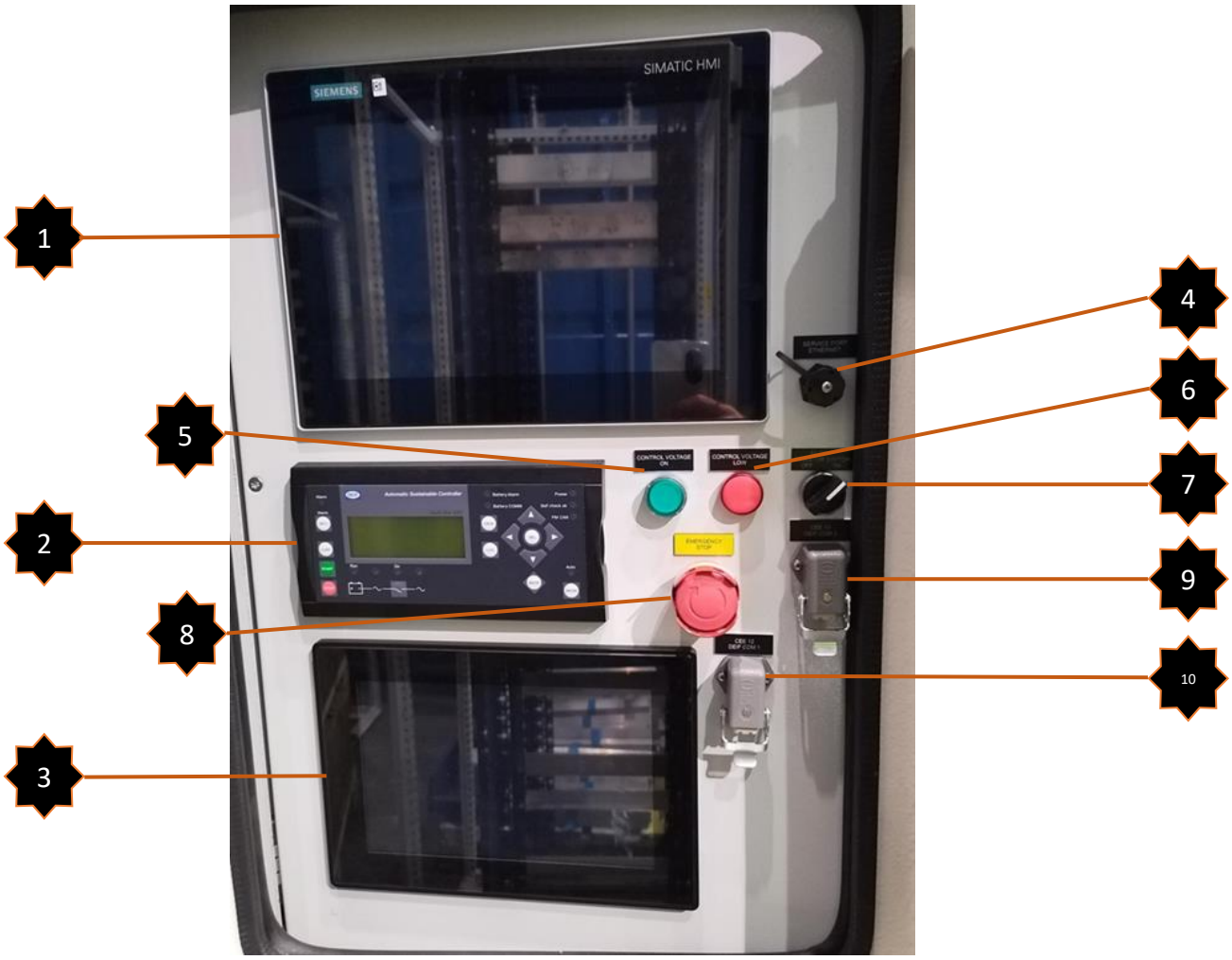


3.1.7 HMI Display & Control Panel

At the front of the container behind the hatch, there are various instruments for displaying, operating and controlling the BQ battery container. The basic functions and relevant information are described below. For a more detailed description of the processes see section 4.







1	HMI – Display Container (PSW)
2	Control Panel - DEIF
3	HMI – DEIF
4	Ethernet Port RJ45 (Service)
5	LED Button – Start Control Voltage from UPS
6	LED Control Voltage “Low”
7	Switch for Deif / CAN-Bus end of line resistor
8	Emergency Stop Switch
9	Deif COM (Interface) / CAN-Bus for Power Management
10	Deif COM (Interface) / CAN-Bus for Power Management

3.1.8 Internal Control Voltage 230V

The voltage control of the system can be supplied in several different ways.

#1 – External Storage Supply (CEE02 -230V/50Hz) is connected.
If a 230V storage supply is available, the control system will be fed by this as long as the battery system is stopped. When the start button on the HMI is pressed, the system will switch to internal supply of the control voltage. The system takes about 45 seconds to start producing power from the internal battery bank. In this time period the 230V control voltage will not be available. The light inside the container will therefore not be powered for these 45 seconds. During this time the 24V control system is supplied from the UPS battery. When the startup process is finished, all the control voltage will be fed from the internal battery racks.

#2 – Without External Power Supply– with Operating Input Drive
If a 230V storage supply is not available, the control system will be powered by the UPS battery. The UPS starts with the "Control Voltage On" button. The control system takes about 90 seconds to power up. After the control system is powered up, the system can be started with the "Start" button on the HMI. After the startup process is finished, all the control voltage will be fed from the internal battery racks/strings. If the battery system is not started within 15 minutes after the UPS is started, the control system will be shut down. A fully charged UPS has about 40 minutes of available energy (2.5 cycles).

#3 – Input Supply Power locks or CEE 125A (Grid / Generator) is connected
If there is a feed connection at the input power locks and the selector switch QB2 is in “charge” position or at the CCE01 and the switch QB1 is on, the control voltage can be fed through this supply. This will happen automatically if the control system is started. If the control system is not on, it will have to be powered up from the UPS battery by pressing the "Control Voltage On" button. After the control system is started, the control voltage will be fed from the 400V input connection.

#4 – Without Power Supply from External – with Operating Output Drive
Read the Chapter “Energy Saving Mode” (=> 3.1.10)

In the moments when the system switches between the described sequences, there will be for a short time a blinking from the illumination inside the container. This is a normal part of the operation of the battery system.

3.1.9 Feedback from the VFD Main breaker (Door switch)

The main breaker / switch [QB4.1] in front of the door (VFD cabinet) must be in the ON position. This provides a feedback signal from the switch to the PLC.



3.1.10 Energy Saving Mode

To decrease system energy loss, hardware and software have been put in place to shut down the input module when it is not needed for operation. This energy saving control makes it possible to shut down the input module.

Procedure to activate it:

1. The shutdown of the input module will be done by a **timer**.
    a. In the menu "Energy save mode" on the HMI is an "Activate" button and a time which could be set from the customer. If this "Activate" button is green, then the function is selected. In the moment when the output module is running, the "Time to Activation" will count down. After that the input module will stop automatically. Now the reefer and the internal control voltage will be fed through the output module.

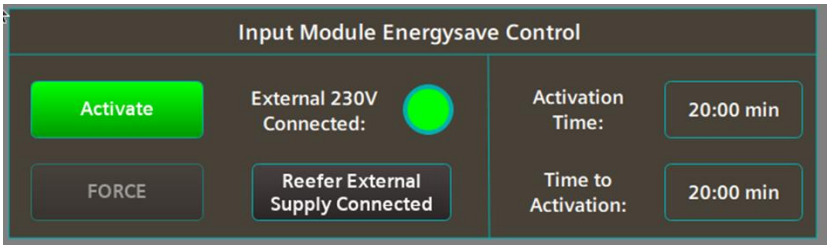
2. The shutdown of the input module will be done **directly** by the **"Force"** button.
    a. If there is no "external 230V connected" to the CEE02 socket:
        - Verify that the output module is in operating mode (running)
        - Press the "FORCE" button
        - The input module will shut down directly (~2-10 sec.)

3. If there is "external 230V connected" (CEE02), the plug from the reefer is connected to an external supply and the output module is not running:
        - Verify, that the reefer unit is connected to external 400V and is running
        - To confirm that, click on the HMI on the button "Reefer External Supply Connected", this button will be green
        - Press the "FORCE" button
        - The input module will shut down directly (~2-10 sec.)

        ⇨ This will keep the DC-link powered without any of the converters running.

Procedure to deactivate it:
If the external 230V is disconnected and the output module is not running, the input module will start. Also, when the input module is started again, the "Reefer External Supply Connected" button will be reset for safety reasons. If a grid is detected on the input, the input module will automatically switch to "ON-GRID" mode to charge the unit. The Activate button makes it possible to choose whether the Input Module Energy save control is activated or deactivated.

The following interface is accessible in the PON Admin menu:



3.1.11 Internal Shutdown Sequences

Shutdown caused by missing battery string

The battery strings need to be within +/- 5V to be allowed to be connected together during startup.
To prevent strings getting out of balance a shutdown function is put in place.
If a string is disconnected while the battery system is running, the shutdown function will shut down the battery system after 1 minute.

Shutdown fault input module

If the system is running in off-grid mode and energy save mode is not active, the internal control voltage and cooling is supplied from the input module.
A fault in the input module would therefore lead to overheating of the system, and the control voltage would also trip when the UPS has run without charging for 15 minutes.
Therefore, a function that shuts down the system if a fault occurs in the input module has been implemented. If the input module is in fault state for 10 minutes the system will shut down.
If the fault is cleared and the "Start" button in the HMI is pressed, the input module will start again, and the battery system will continue in normal operation.

3.1.12 External / Remote Access (Cloud & Service)

It is possible to establish a remote connection to the internal network via GSM modem (VPN) and control the system for a short time. However, the connection must not be maintained as the data bandwidth can be exceeded (monthly quota depends on SIM card and provider).
In addition, there is a datalogger PC inside the control cabinet. This is connected to the PSW Cloud Data Server and sends data cyclically. For more information, see the separate manual for the PSW Cloud Solution.

3.1.13 GPS Location Device

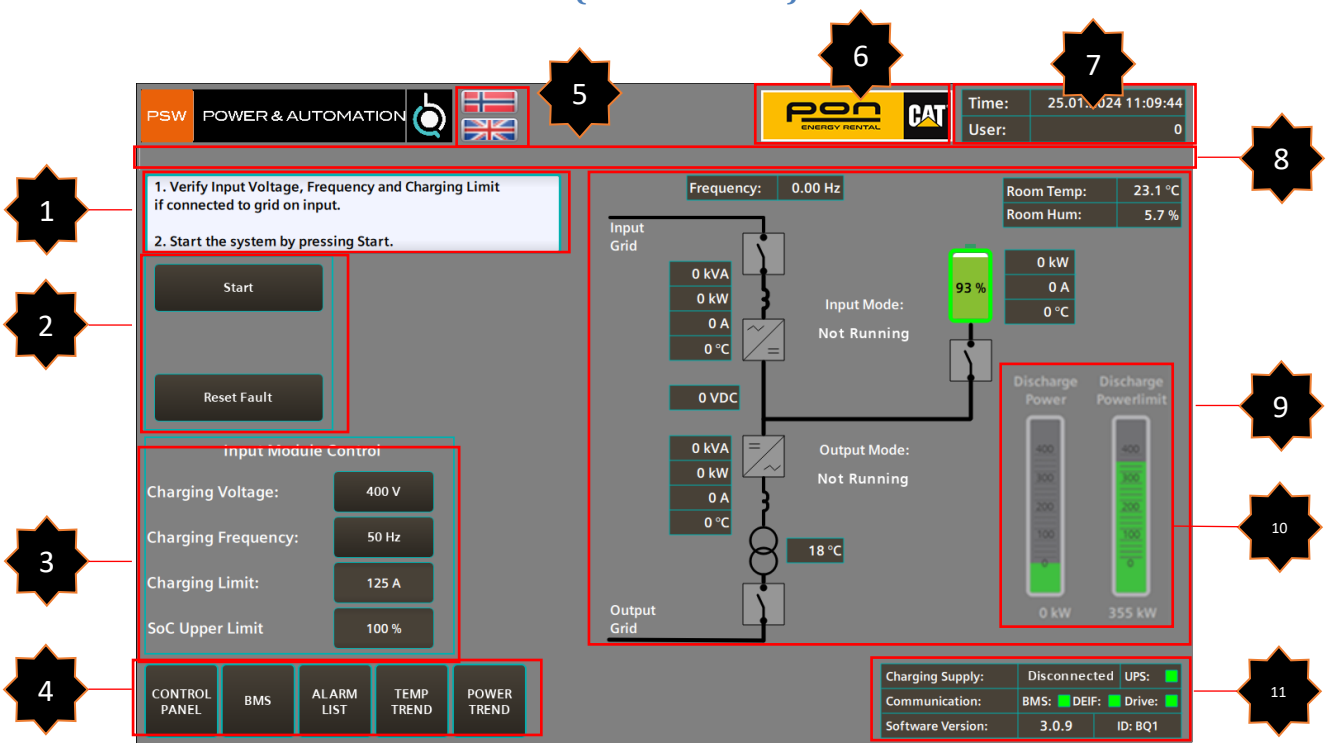
A GPS location device is installed in the container, which transmits the current position to the PSW cloud.

3.1.14 Energy Monitoring (Socket Outputs)

Energy monitoring is implemented to measure the energy usage from the socket outputs.
This data is only used for monitoring and is streamed to the cloud where the data is stored.

3.2 HMI Display Container (PSW)

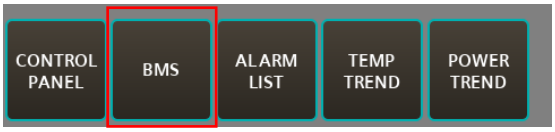
3.2.1 Main Overview (Control Panel)

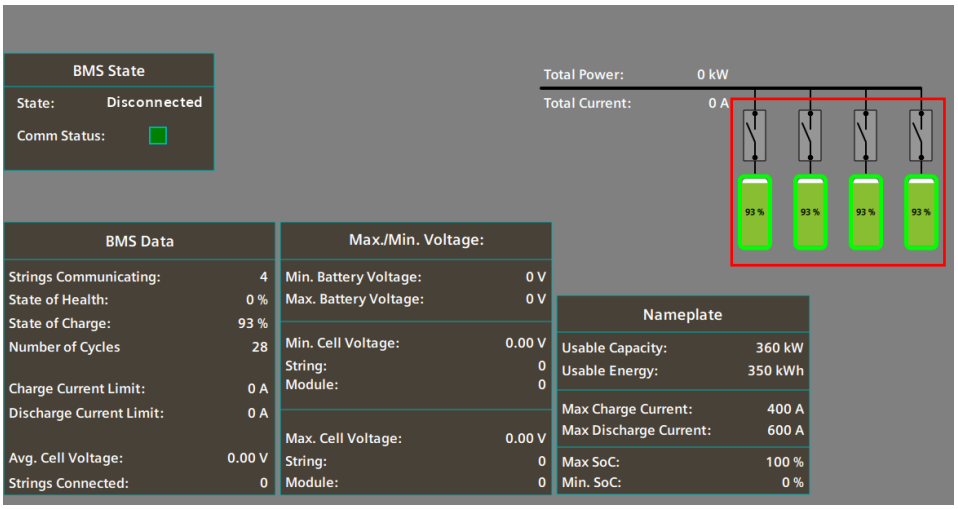


1	Text status bar, shows the actual sequence and what needs to be done
2	Buttons for starting and stopping the battery system. Reset button for resetting faults. Start and stop buttons are only visible if start/stop is allowed.
3	Charging setting options for voltage, frequency, SoC and current limit (only during “on-grid” connection)
4	Navigation, switch to other views (BMS, alarm list and trends)
5	Selection languages
6	Customer Logo “button”, press for advanced settings (service level)
7	Display date, time and logged in user
8	Status bar for alarms and warnings. In case of no faults, the bar is not visible. Alarms and faults are displayed in red and warnings in yellow (for details => section 7.2).
9	Main area for the status of the BQ container including the values and current flow direction. During charging, the arrows are displayed in blue and in case of discharge or conversion in green.
10	Status bar – discharge power and limit (only visible in running mode)
11	Status bar for the network connection between the different components (PLC, FQ drives, BMS, Deif)

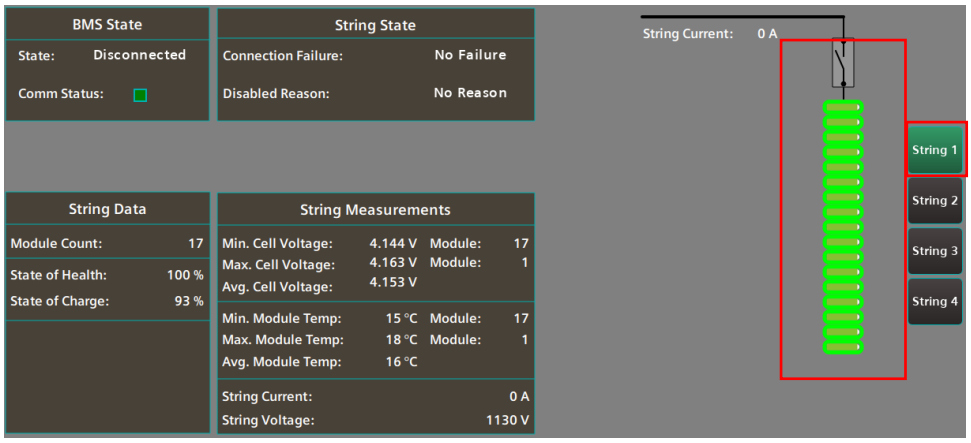
3.2.2 BMS (Battery Management System)

To get more details about the battery system, click on the "BMS" button in the navigation. The following view gives you an insight into the status of the 4 individual racks.





If you click on one of the strings, you will be directed to the corresponding rack.



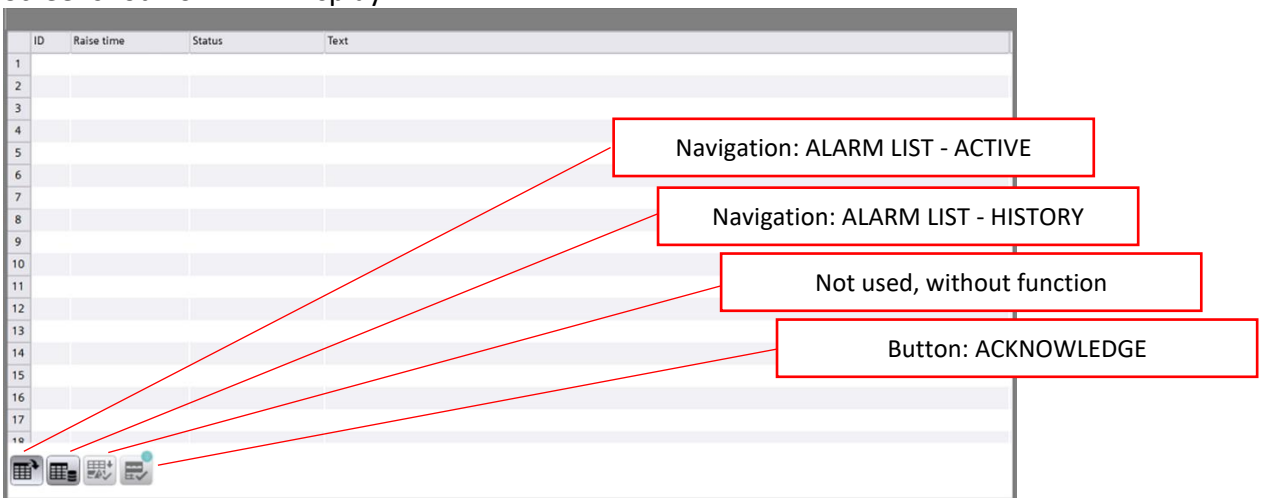
3.2.3 Alarm list and alarm history

Click on the "ALARM LIST" button in the navigation. In the normal operating state, there are no alarms or warnings in the table. If you want to see the alarm history, press the appropriate navigation button (see image below). If there is an active alarm or error, you can use the acknowledge button.

Declaration:

Definition	Color (active)	Color (ackn.)	Description	Comment
Alarm	red	orange	This can, for example, be a fault in bus communication or a message from a communication peripheral unit of the container. These messages are relevant to the system and can lead to the battery system being switched off. As soon as the alarm has "gone" or " been resolved", it must be acknowledged. This message is then no longer displayed in the active window but is shown in the history as a past message.	
Fault	red	orange	This can be a critical fault, such as a motor protection switch tripping or something similar. These messages are relevant to the system and can lead to the battery system being switched off. As soon as the alarm has "gone" or "been resolved", it must be acknowledged. This message is then no longer displayed in the active window but is shown in the history as a past message.	
Warning	yellow	-	A warning is an indication that something must be checked. These messages are not necessarily directly relevant to the system and do not impair the function in any essential way. This type of message does not have to be acknowledged, after it has "gone" or "been resolved", it is automatically displayed in the history.	

Screenshot from HMI Display:





If you have an incoming fault or alarm, it will be displayed anywhere on the HMI display at the top.
This message is displayed in red in the alarm table as an active message. In this example, an emergency stop was activated.



As soon as the alarm or fault has been resolved and is no longer active, it is still displayed in orange in the same table. Only after pressing the acknowledge button will this message no longer be displayed in the active window and will be added to the history.



If there is a multiple number of active alarms or warnings, the "newest" is always displayed at the top of the table. If there is an active warning, e.g., when measured values exceed established limits, these are displayed in yellow. After this warning is no longer active, it is automatically removed from this list and then displayed in the history. In the alarm list history all alarms, errors and warnings are displayed. These are presented in chronological order from "newest" to "oldest" (top to bottom).



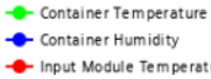
Check and look at the history regularly to know if there have been any messages while you were away.

3.2.4 Temperature “Trend” & “Trend Log”

In the back of the HMI screen, there is a memory card which saves logged data. The capacity depends upon the number of parameters logged and the rate of changes. Typically, 3-5 days can be stored.

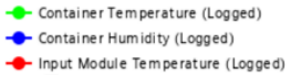
<u>Switch between the two views as follows (“live” + “logdata”):</u>

Press 

, you will see the “live” diagram 

/ button change to 

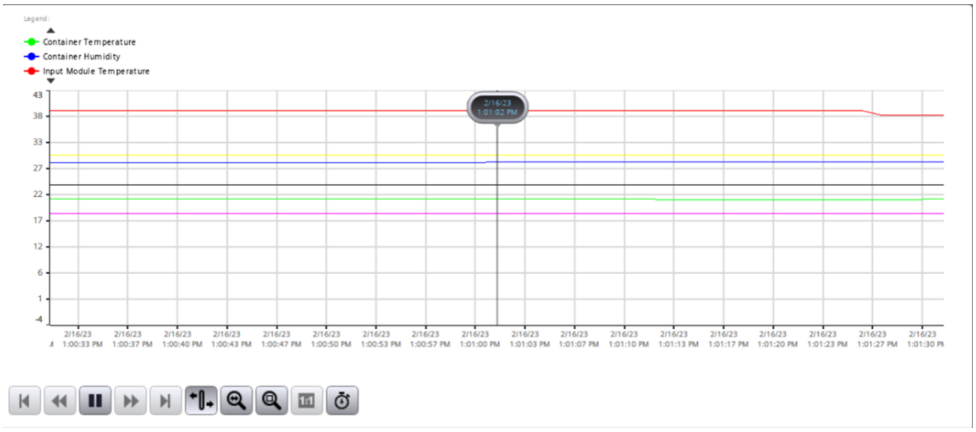
Press 

, you will see the “log” diagram 

/ button change to 

Press on the navigation button “TEMP TREND LOG”. Here you can view values in a line diagram.
The measured values shown here are:

- Container Temperature (green)
- Container Humidity (blue)
- Input Module Temperature (red)
- Output Module Temperature (yellow)
- Trafo Temperature (pink)
- Max Output Module Temperature Battery (black)



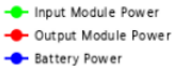
If you press the button “break / pause” you can slide the timeline to view the values.

3.2.5 Power "Trend" & "Trend Log"

There is a memory card in the back of the HMI screen which saves logged data. The capacity depends upon the number of parameters logged and the rate of changes. Typically, 3-5 days can be stored.

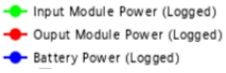
Switch between the two views as follows ("live" + "logdata"):

Press 

, you will see the "live" diagram 

/ button change to 

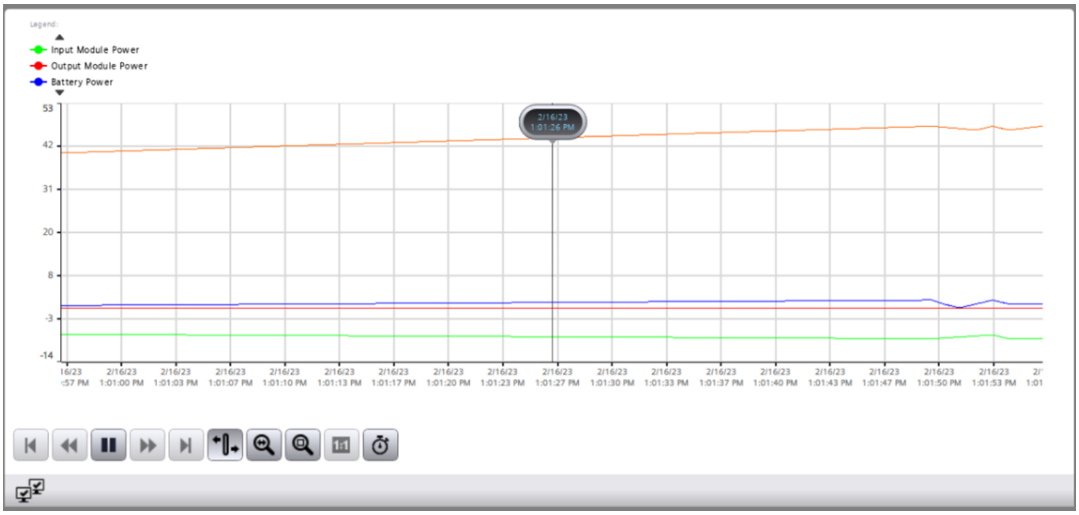
Press 

, you will see the "log" diagram 

/ button change to 

Press the navigation button "POWER TREND LOG". Here you can view values in a line diagram.
The measured values shown here are:

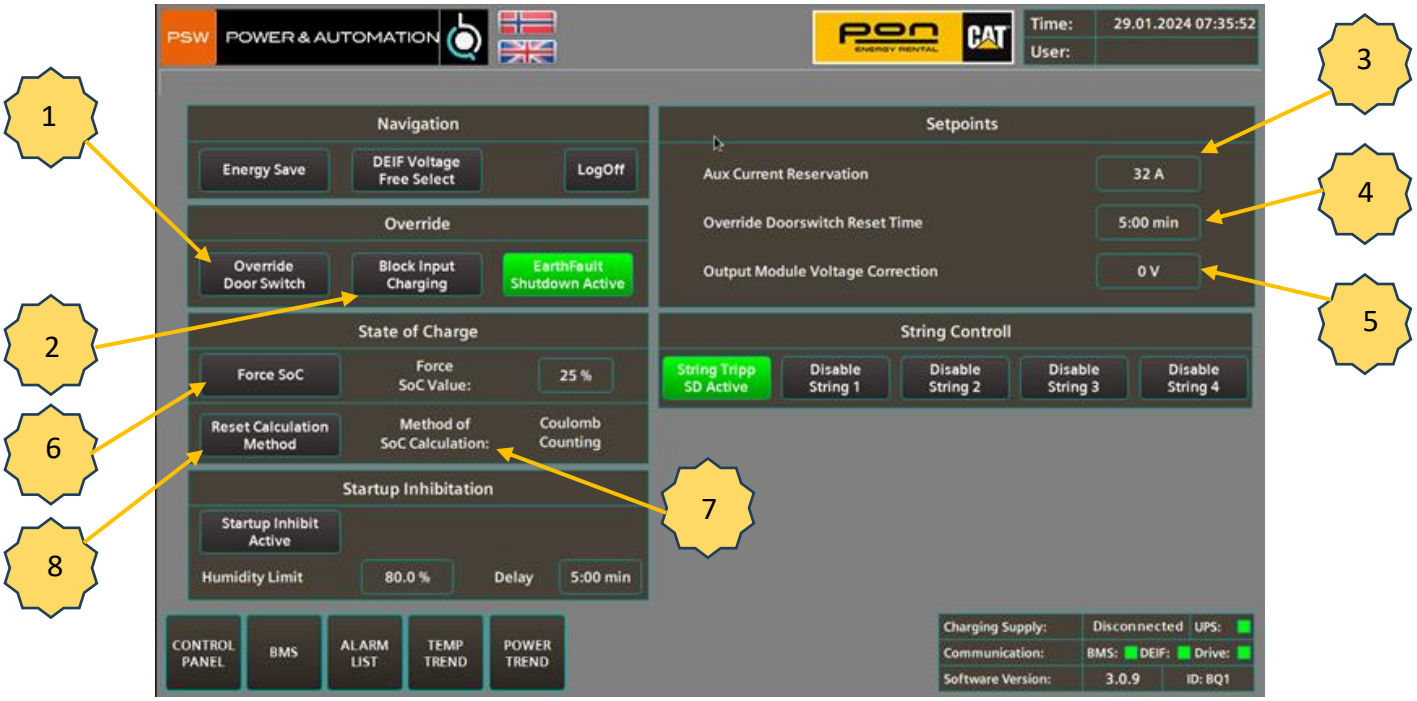
- Input Module Power (green)
- Output Module Power (red)
- Battery Power (blue)
- Battery SoC (orange)



If you press the button "break / pause" you can slide the timeline to view the values.

3.2.6 Customers Settings

To get to customer settings, press your own company logo on the HMI display (PSW).
Login with your username and password.



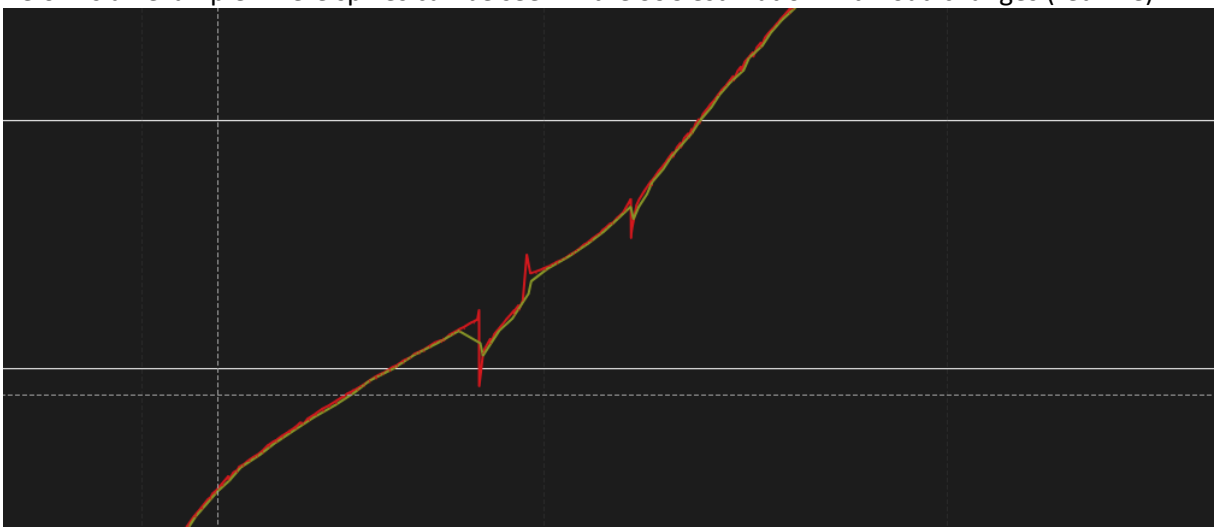
1	Override Door Switch	If this button is activated, the container door can be opened in operating mode.
2	Block Input Charging	This button locks the input module from switch to on-grid mode. It can be deactivated by the operator if charging from the input is needed.
3	Aux Current Reservation	Reservation of power requirements for auxiliary components as a buffer.
4	Override Door Switch Reset Time	Time to reset the door switch override. After the door has been opened and closed, the time begins to count down. When the set time has reached 0, the override is deactivated. This is to ensure that the door switch resets to active.
5	Output Module Voltage Correction	This is used to make an offset in the voltage setpoint, which is sent to the output module. This can be used to balance reactive loads between the battery system and gensets when running un parallel.
6	Force SoC	Thes will force the Soc value of the system to use a selected value. This can for example be used to get the unit running if it has been fully discharged and charging is only available through the output module.
7	Method of SoC Calculation	This status shows which SoC calculation method is used to calculate the SoC (%) Coloumb Counting: Kore Power SoC (%) OCV: PSW P&A SoC (%)
8	Reset SoC Calibration	This button will change the SoC calculation method from OCV to coulomb counting, if the coulomb counting value is within 10% of the OCV value.

Details and Information for the SoC Calibration Alarm & Calculation:

Inside the PLC application from PSW is a function, which compare the current “state of charge” from the KPMC and an own calculation method, based on “Open Circuit Voltage”.
This is a calculation based on the correlation between open circuit voltage and SoC.
The cell voltage varies with load and the algorithm can be used in two different ways.

- The pro of this algorithm is that it will not drift off over time.
- The con of this algorithm is that it will fluctuate with load changes.

Below is an example where spikes can be seen in the SoC estimation with load changes (red line):



Inside the application there is an estimate of the SoC with a combination of these two algorithms. The “Coulomb Counting” method will be the main method of calculation, but if the “Coulomb Counting” value differs >15% from the “OCV” method, the system will change over to the “OCV” as the method of calculation.

If deviation is detected (Coulomb Counting values changes more than 5% from a cycle to another), and the coulomb counting value differs <15% from the “OCV” value, the system will change back to using the “Coulomb Counting” value as the method of calculation.

The operator can also change back to the “Coulomb Counting” method by pressing the “Reset SoC Calibration Alarm” button in the PON menu. This will only work if the coulomb counting value differs <15% from the OCV value.

3.2.7 Description of Discharge and Charge Limit

Battery current limits
The batteries in the BQ system are monitored by the KPMC unit inside the control cabinet. The KPMC is the main controller of the battery system and communicates with the RBMS units on top of each battery rack, which then communicates with each module in the battery rack.

The KPMC communicates the charge and discharge limits to the BQ control system. These limits are communicated to avoid overcharging or discharging of the batteries. The limits communicated are calculated based on the highest/lowest cell voltage in the system at any given time.

The discharging limit calculations are based upon the following table:

Current Limit Batteries	Power Limit Discharge	Cell Voltage
600 A	360 kW	> 3.4 V
150 A	250 kW	3.4 V – 3.2 V
120 A	100 kW	3.2 V – 2.98 V
60 A	45 kW	2.98 V – 2.9 V
0 A	0 kW	< 2.9 V

The charging limit calculations are based upon the following table:

Current Limit Batteries	Cell Voltage
400 A	< 4 V
300 A	4 V – 4.2 V
200 A	4.2 V – 4.28 V
60 A	4.28 V – 4.295 V
10 A	4.295 V – 4.298 V
20 A	4.298 V – 4.325 V

Note that these tables are based upon the battery current, at the DC link with a voltage of greater than 800V. The actual current limits on the grid side of the frequency converter will be higher than these limits and will therefore be restricted by other factors than battery discharge and charging limits.
The BQ control system automatically takes these limits into consideration when controlling the frequency converter input and output module. The different scenarios that system can encounter are:
Charging through input module:
When charging though the input module, the limits will be taken into consideration, and the charging current when approaching 100%, SoC will automatically reduce.
Charging though output module:
When charging though the output module, the limits will be taken into consideration, and the charging current when approaching 100%, SoC will automatically reduce.

Discharging though output module:

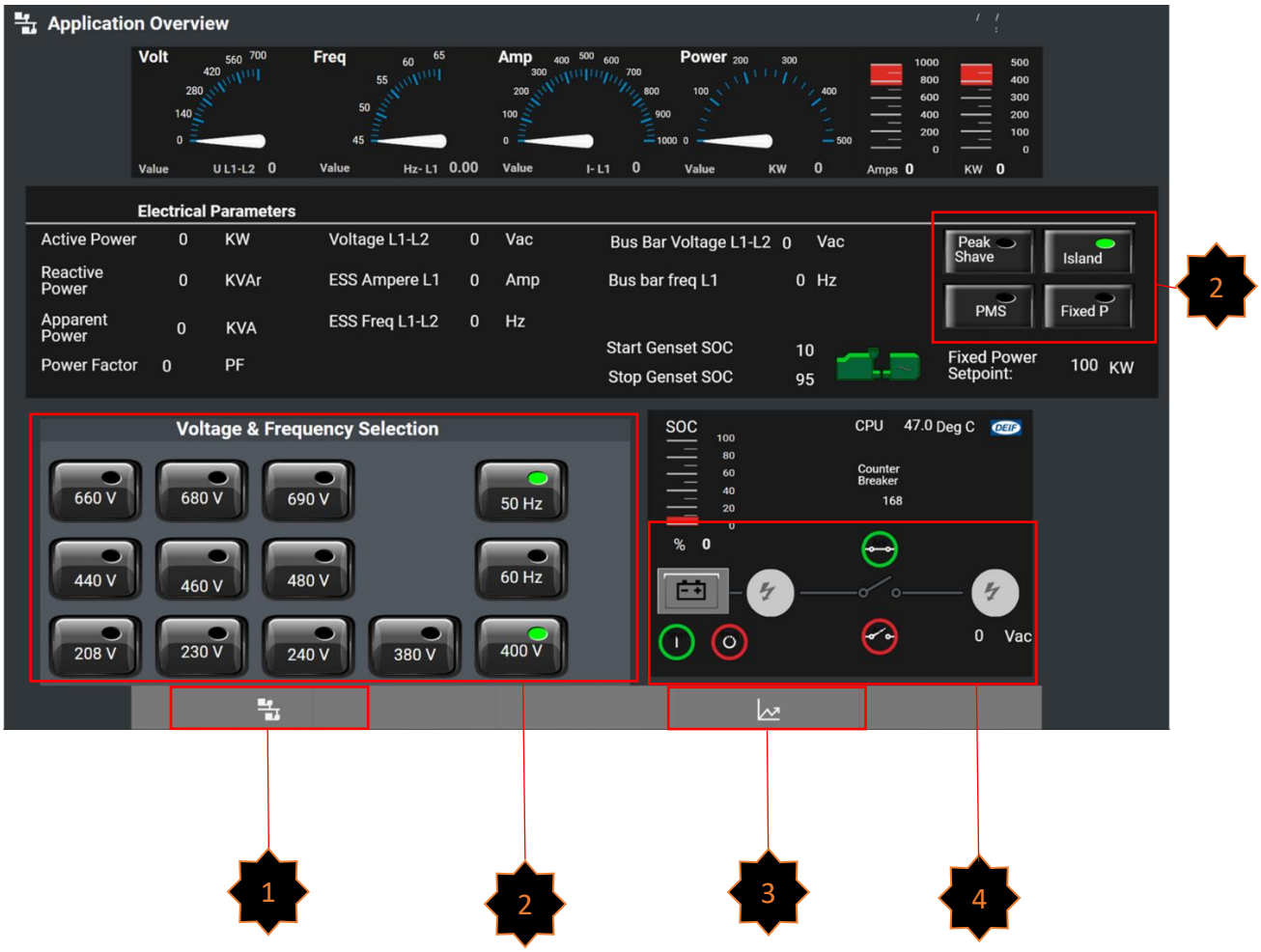
If a discharge limit is exceeded the output module will try to drop load by reducing the frequency. The load will then have to be supplied from other sources on the grid.
If the BQ system is the only power source on the grid, the consumer load will have to be reduced to avoid a trip. If consumer load is not reduced, the frequency of the grid will be reduced, and the system will trip due to low frequency.
To avoid this, the BQ system is monitoring the power limit in the upper HMI display when output module is providing power to the connected grid.



The "Discharge Power" bar indicates how much power the output module is discharging the batteries.
The "Discharge Power Limit" bar indicates the discharge power limit based on the lowest cell voltage in the battery system. When the lowest cell voltage hits 3.5 V the "Discharge Power Limit" bar will gradually decrease and hit 250 kW when the lowest cell voltage hits 3.4V. The actual limit communicated to the frequency converter does not come into effect until the lowest cell voltage hits 3.4 V. This is to allow the operator to reduce load on consumer side before the limit is reached to avoid a system trip.
The limit shown will continue to decrease towards 0 kW as the lowest cell voltage decreases. Actual limits communicated to the frequency converter will continue to decrease based on the discharging limit table.
If the batteries are charged from the input module at the same time as the output module is discharging the batteries, this will be taken into consideration for both the shown and the actual limits.

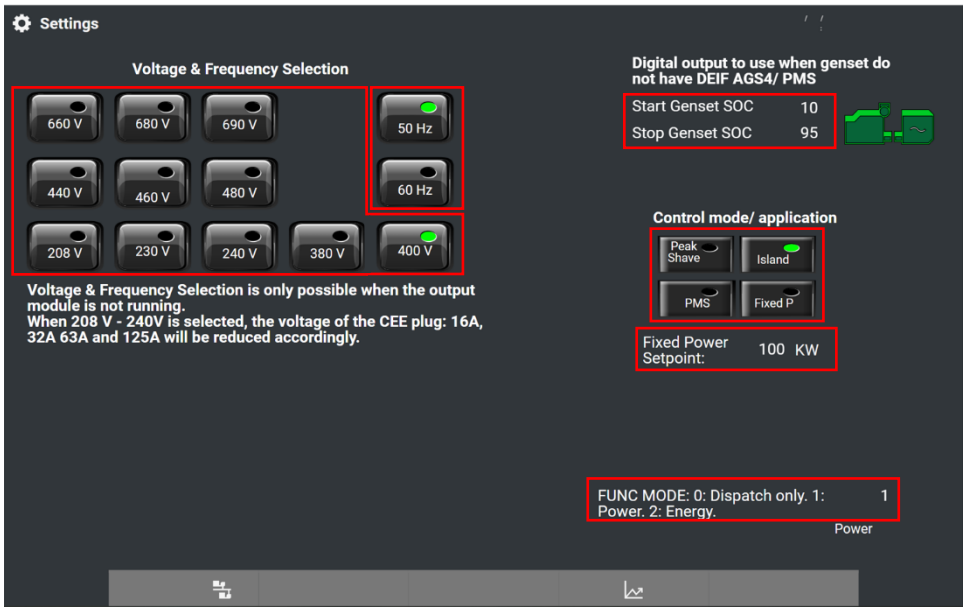
3.3 HMI Display DEIF

This screen is used to display the current values and settings. To change setpoints or operating modes, click in the lower area in the field for selecting the voltage and frequency. A new window will open with the possible settings.



1	Navigation to this main screen
2	Navigation to the setting screen
3	Navigation to the trend screen
4	Control Buttons (same function – Deif control panel)

Setting screen:

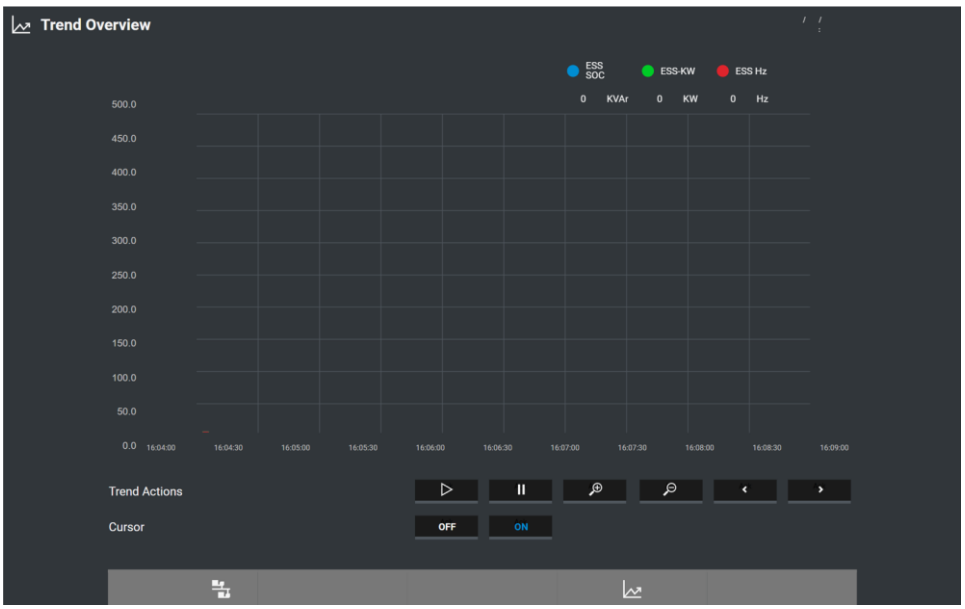


Setting selection:

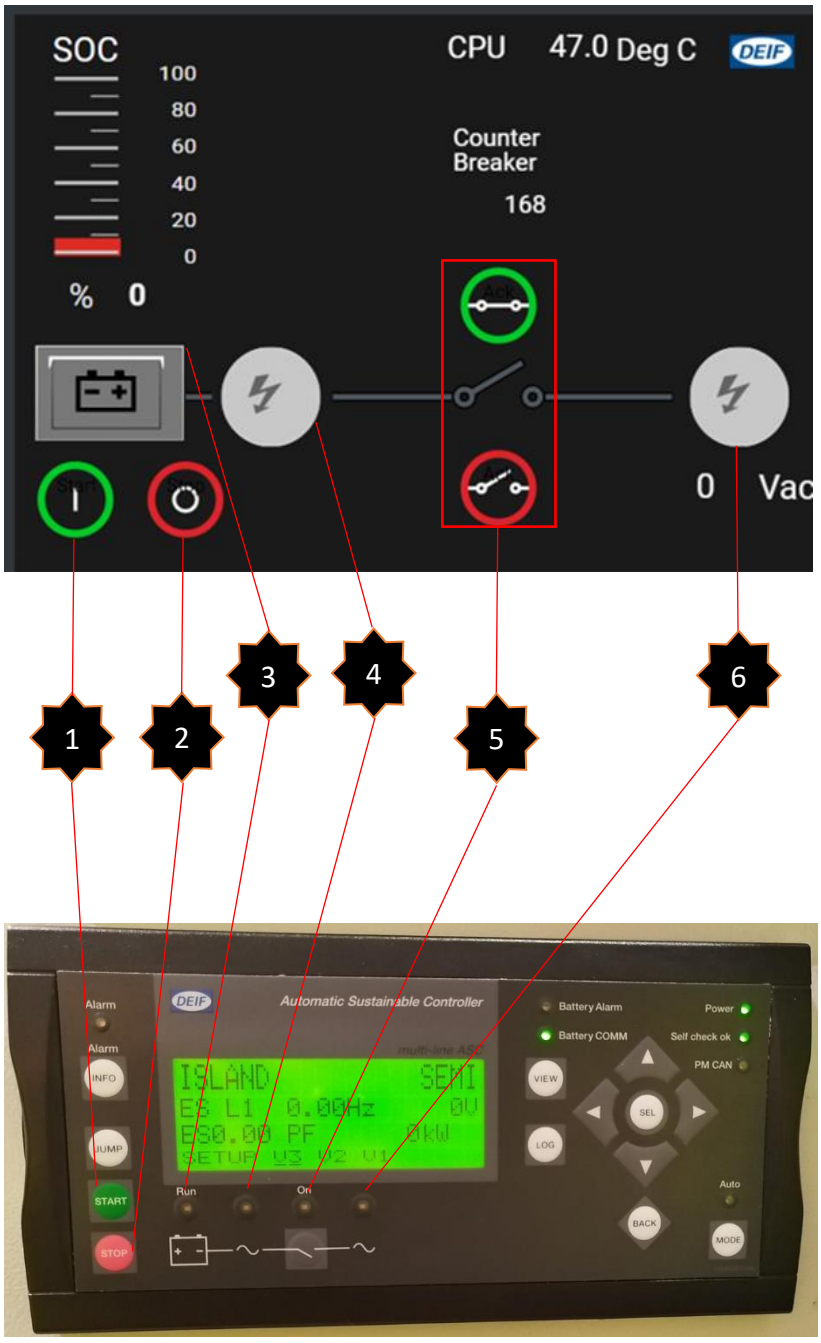
- Voltage
- Frequency
- Start/Stop Genset SOC
- Control mode
- Fixed Power
- Function Mode

When you are changing settings, you must verify voltage and frequency before starting.

Trend overview screen:



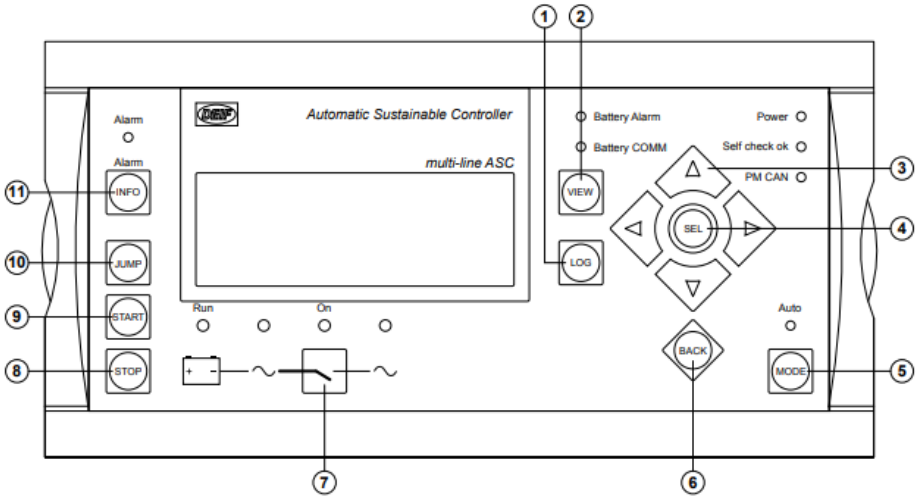
Control buttons on DEIF HMI:
You can either use the control panel or the DEIF HMI to operate the output module.



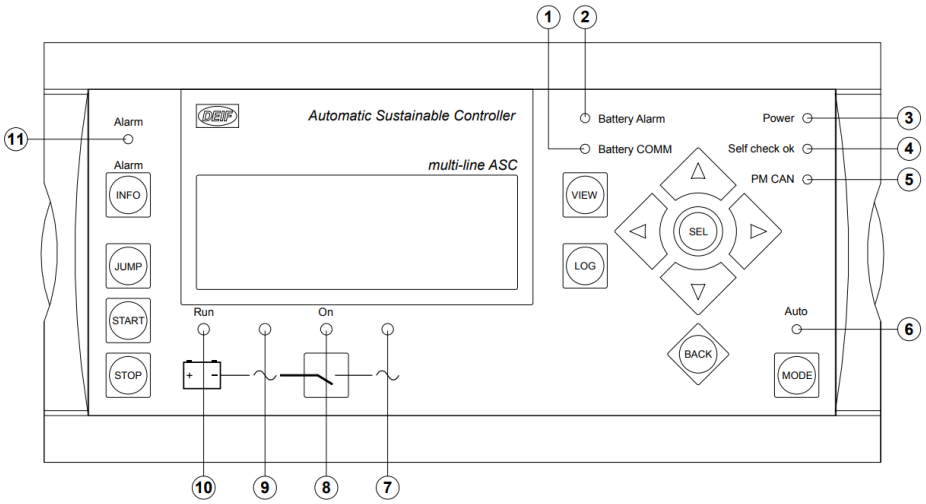
No	Function
1	Start
2	Stop
3	Status indication “running”
4	Source busbar status
5	Output breaker (on/off)
6	Output busbar status

3.4 Control Panel DEIF

Use the separate user manual for using the Deif Control Panel ASC-4.



1	LOG	Displays the LOG SETUP window. You can view Event or Alarm logs. The logs are not deleted when the auxiliary supply is switched off.
2	VIEW	Shifts the first display line in the setup menus. No function in the View screen (V1-V2-V3). When more than one display is connected, push for two seconds to change the display to the master display.
3	UP/DOWN/ LEFT/RIGHT	Use up, down, left and right for navigating in the display unit.
4	SEL	Selects the underscored entry in the fourth line of the display.
5	MODE	Opens the mode selection menu to choose between AUTO and SEMI mode.
6	BACK	Go one step backwards in the menu (to the previous display or entry window).
7	ESB On/Off	Breaker is open: Press to start the close breaker sequence (if SEMI mode is selected). Breaker is closed: Press to start the open breaker sequence (if SEMI mode is selected).
8	STOP	Stops the ESS (if SEMI mode is selected).
9	START	Starts the ESS (if SEMI mode is selected).
10	JUMP	Enter a menu number to select and display any setting without having to navigate through the menus.



1	Battery COMM	Red (flashing): Communication with the ESS is faulty. Off: There is no communication with the ESS.
2	Battery Alarm	Red (flashing): Alarms are present on the ESS. Go to the ESS to see which alarms are present. Off: There are no ESS alarms.
3	Power	Green: The power supply is on.
4	Self check OK	Green: The ASC is OK (microprocessor supervision).
5	PM CAN	Green: The power management CAN bus is working without faults. Yellow: Power management is not selected. Red: There is a fault on the power management CAN bus.
6	Controller mode	Green: AUTO mode is selected. Off: SEMI mode is selected.
7	Busbar status	Green: Busbar voltage and frequency is OK. Off: There is no busbar voltage.
8	Breaker status	Green: ESS breaker is closed. Yellow (flashing): ESS breaker load time has not expired. Off: ESS breaker is open.
9	Source busbar status	Green: ESS voltage and frequency is OK. Off: There is no ESS voltage.
10	Source status	Green: ESS converter is running.
		Off: ESS converter is not running.
11	Alarm status	Red (flashing): Unacknowledged alarms are present on the controller. Red: All alarms acknowledged, but alarms are present on the controller. Off: No alarms are present on the controller.

3.5 Overview of IP Addresses and Login Data

IP Address	Component	Username	Password
192.168.1.1	PLC (Siemens)		
192.168.1.2	HMI Display Container (PSW)		
192.168.1.3	Input Module Drive		
192.168.1.4	Output module drive		
192.168.1.5	Deif Controller		
192.168.1.6	Kore KPMC		
192.168.1.7	HMI Display DEIF		
192.168.1.8	Bender Isolation Monitor (Option)		
192.168.1.12	GPS Device		
192.168.1.100	Rex (remote access for support)		
192.168.1.200 to 192.168.1.220	Reserved for customer IP address		

If connection by laptop/computer is required, verify the local IP settings. There is a reserved IP address range for the customer from 192.168.1.200 to 192.168.1.220.

4 Operational Procedures

4.1 Operational Mode – Cold storage

If the unit is to be stored as “cold storage” the standby supply must be connected. The standby system charges the 24V UPS and powers the de-humidifier.

⚠	WARNING Leaving the control voltage over time will drain the high voltage batteries due to balancing. Always check batteries “State of charge” regularly

1. Connect the standby supply 208-240VAC, 50-60Hz, 1x16A on the outside of the unit.



2. Verify that the de-humidifier is operational by checking the drain outside the unit.





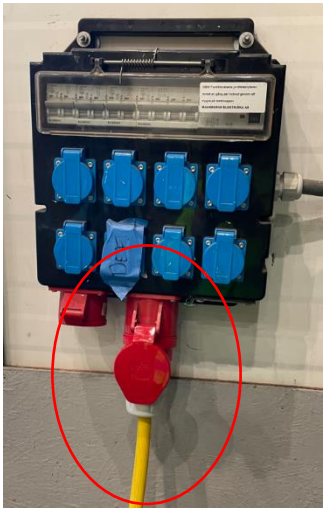
	INFORMATION Cold storage mode shall only be used if the BQ is stored in an environmentally controlled area with temperatures between 0 – 30 °C. If this is not the case the cooling and heating units must always be connected.

If the unit is to be stored as “cold storage” in an area where the temperature is outside 0 – 30°C the cooling and heating unit must be connected.

3. Disconnect the cooling unit socket from the outlet.



4. Connect an external supply to the cooling and heating unit.



5. Verify that the cooling and heating unit is operational.



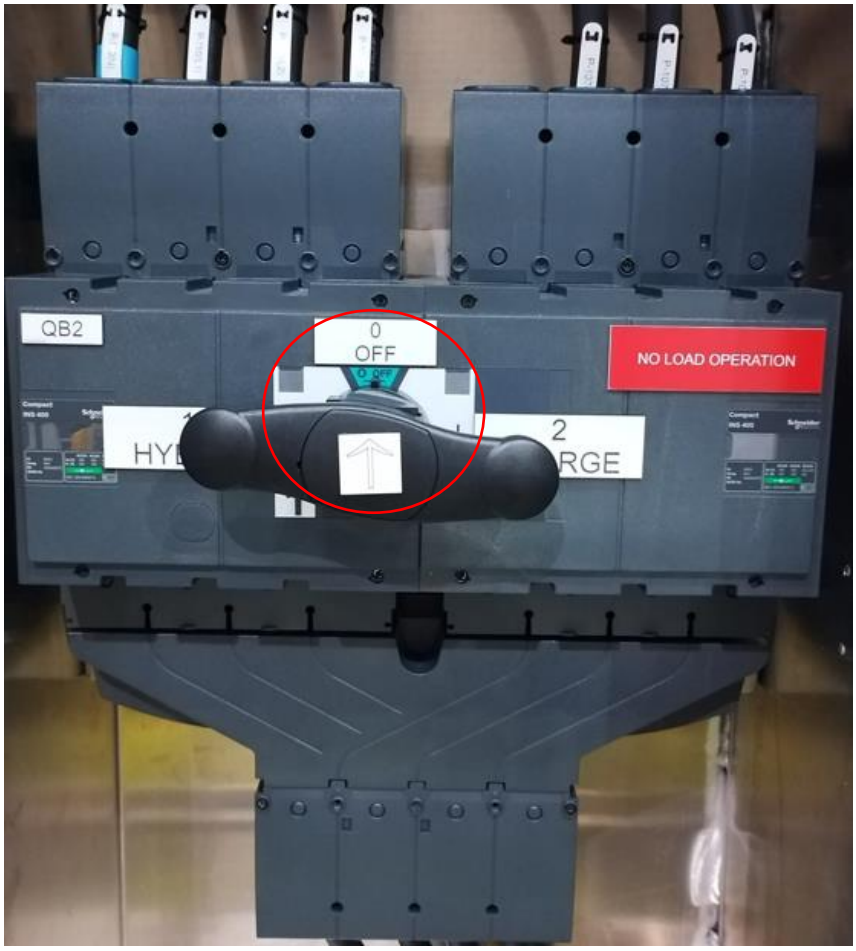
	INFORMATION The cooling & heating unit temperature setpoint must be 18°C. For more information see the cooling & heating user manual

4.2 Input Operation (HMI Display PSW)

4.2.1 Input Module (PSW) "On-Grid"

When a grid power supply of 380-440V, 50-60Hz is available, the unit can be operated in "On-Grid" mode. In "On-Grid" mode the grid power supply is used to feed the internal control system, the cooling & heating units and charge the battery strings.
To setup the BQ for "On-Grid" mode follow the steps below

1. Verify that the CEE Switch (QB1) and the Powerlock selector switch (QB2) is in "0" position in the AC Interface section.



2. Connect the Charge Supply CEE 380-440V 125A or Power Locks in AC input section.



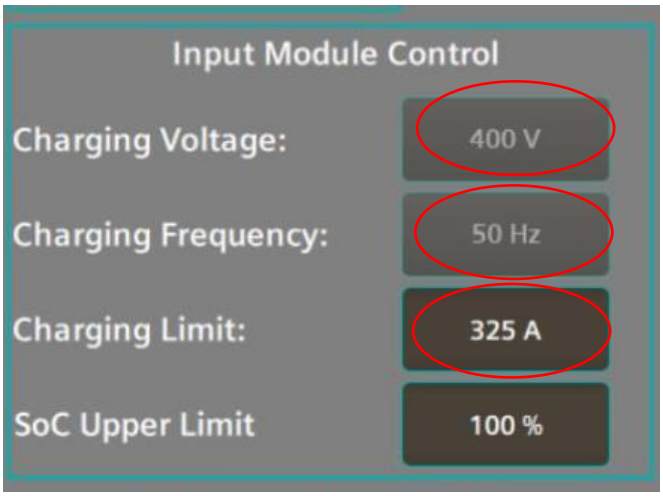


3. Press the 24V control voltage main switch and wait for the PLC to bootup.



4. Make sure that all emergency stop buttons are reset and press reset on the HMI, if the start button is not visible.

5. Set the voltage, frequency and current rating for the input connection.

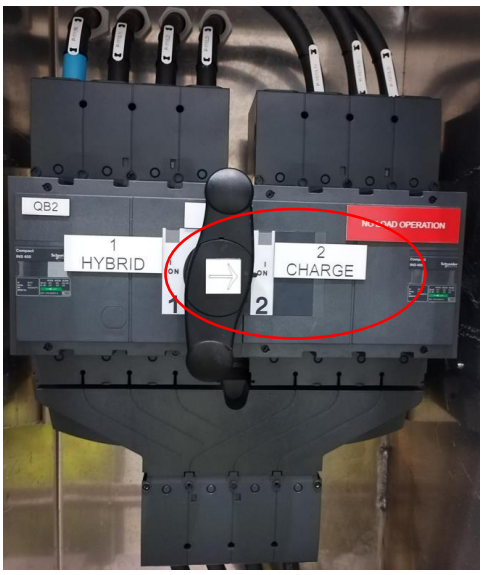


Note: 32A is subtracted from charging limit setpoint to reserve power or the cooler & heater

6. If the Power Lock is connected:
   - Powerlock Selector switch (QB2) - CHARGE
   - CEE01 Switch (125A) Socket (QB1) - OFF

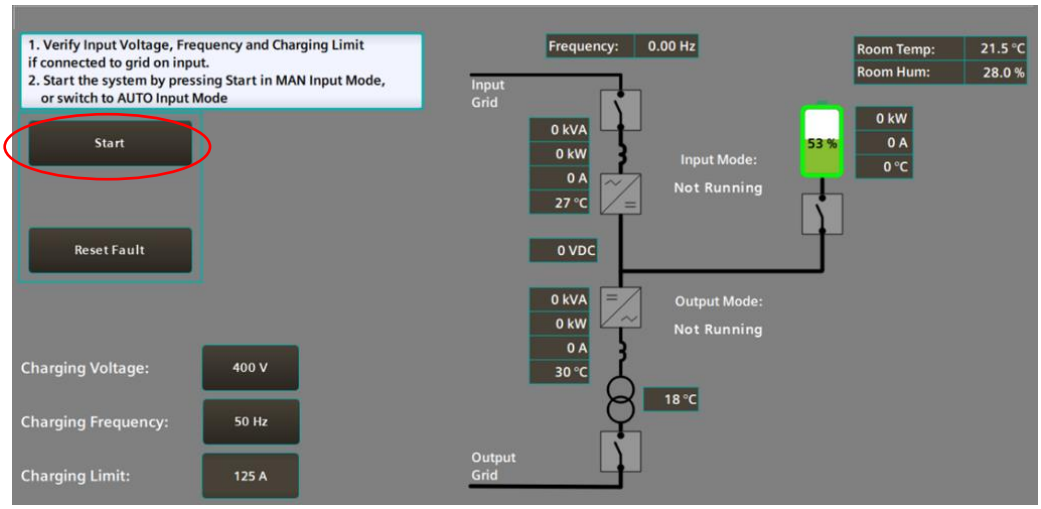
   If the CEE01 (125A) Socket is connected:
   - CEE01 Switch (125A) Socket (QB1) - ON
   - Powerlock Selector switch (QB2) - OFF

⚠	WARNING Don't turn over the QB1 and QB2 when the system is running or a load is connected and operating. These components are marked with red labels "NO LOAD OPERATION". DO NOT turn the QB2 in the "Charge" position and the QB1 in "on" position at the same time.

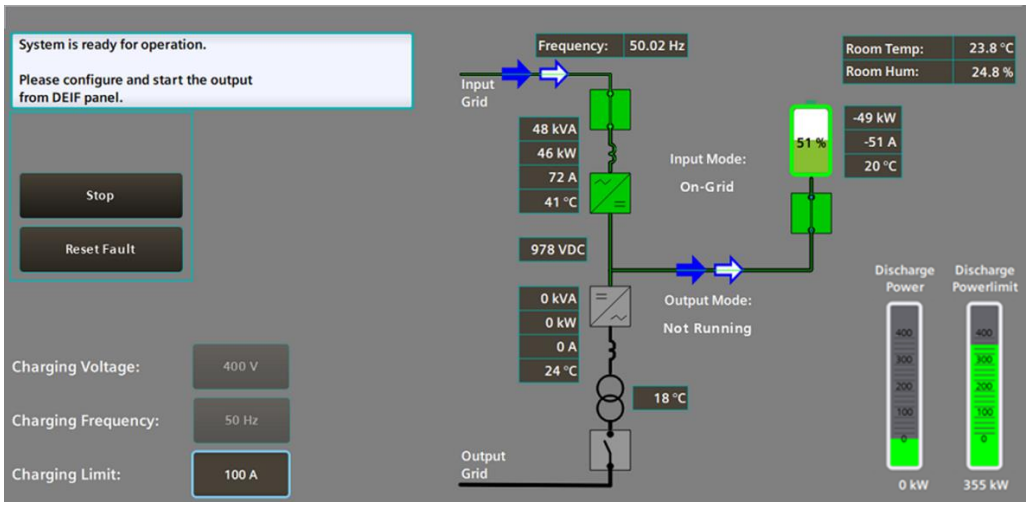




7. In BQ HMI press “start” the input module will start-up



8. Now the BQ-S container is charging the battery and ready for operation in “on-grid” mode. For the next steps you must look at the section “4.3 Output operation” (DEIF).

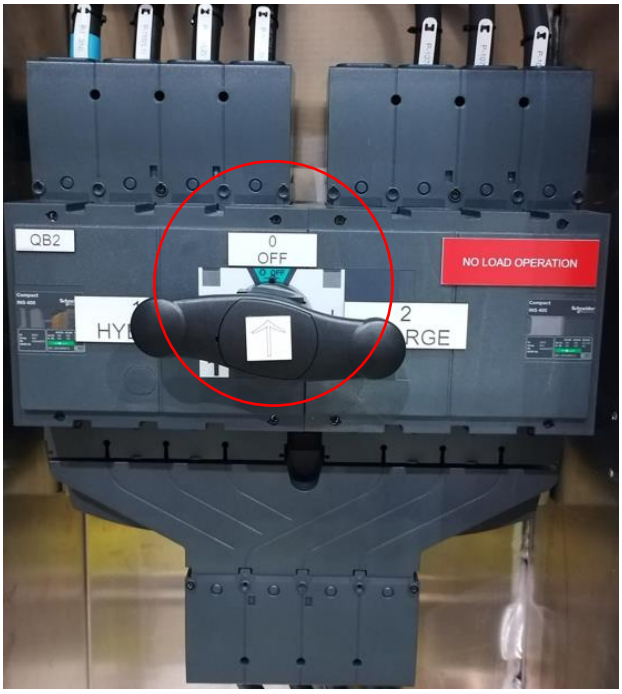


	INFORMATION If the grid power supply is disconnected, the unit will change operational mode to “Off-Grid”. In the transition between “On-Grid” and “Off-Grid” mode the cooling and heating system will be restarted.

4.2.2 Input Module (PSW) / OFF-Grid

When a grid power supply of 400V, 50Hz is not available or the Powerlock selector switch (QB2) is in “hybrid” position, the unit will operate in “OFF-Grid” mode. In “OFF-Grid” mode the internal battery is used to feed the internal control system and the cooling & heating unit.
To setup the BQ for “OFF-Grid” mode follow the steps below

1. Verify that Powerlock selector switch (QB2) and CEE Switch (QB1) are in “0” or “hybrid” position in the AC input section.





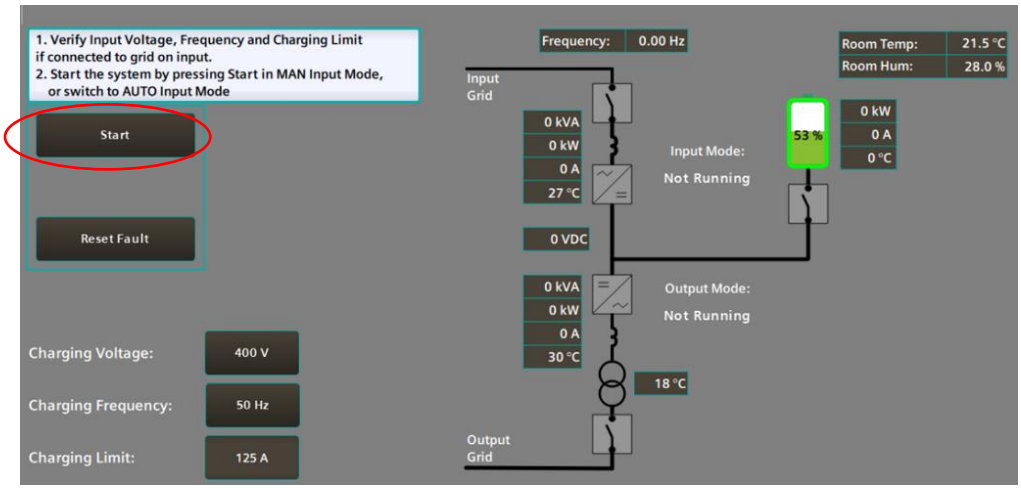
2. Press the 24V control voltage main switch and wait for the PLC to bootup.



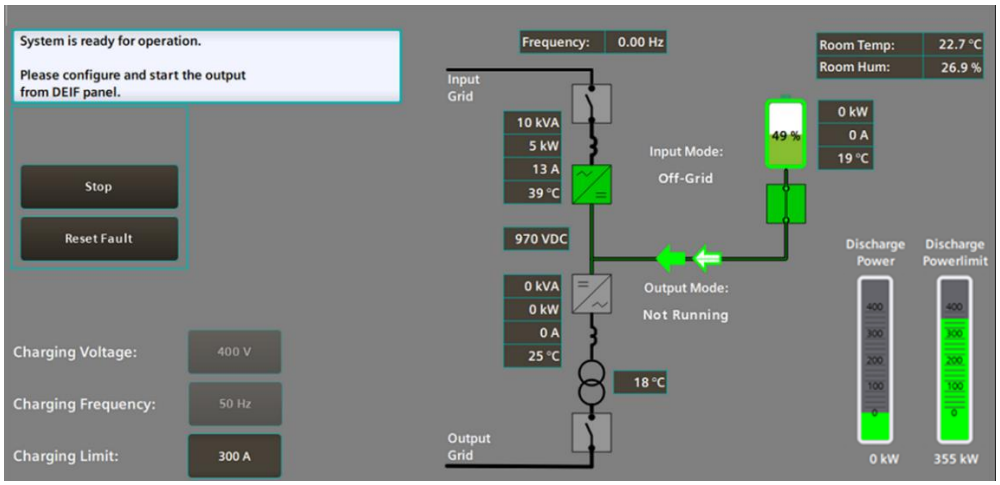
	INFORMATION If no external supply is connected the 24V system will turn off if the unit has not been started within 15 minutes.

3. Make sure that all emergency stop buttons are reset and press reset in the HMI, if the “start” button is not visible.

4. In BQ HMI press “start” the input module will then start-up



5. Now the BQ-S container is ready for operation in “off-grid” mode. For the next steps you must look at the section “4.3 Output operation” (DEIF)



4.2.3 Input Module (PSW) / ON & OFF Grid Operation

The system will change automatically between ON & OFF Grid mode.
If the system is running in OFF Grid and voltage is detected at the input connection, a sequence is started in the control system to switch to ON Grid mode.

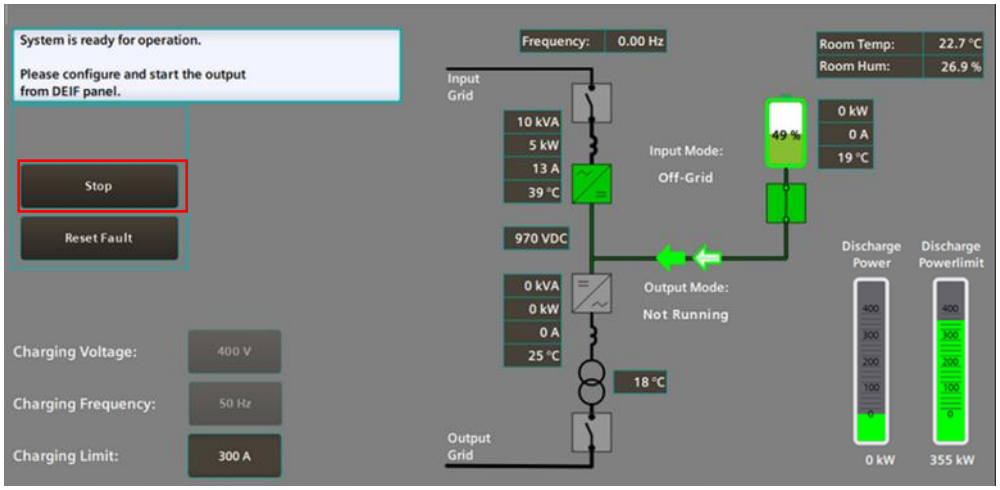
If the system is running in ON Grid mode and the voltage on the input is turned off, a sequence is started to switch to OFF Grid mode.

4.2.4 Input Module (PSW) / Stop Sequence (Shutdown)

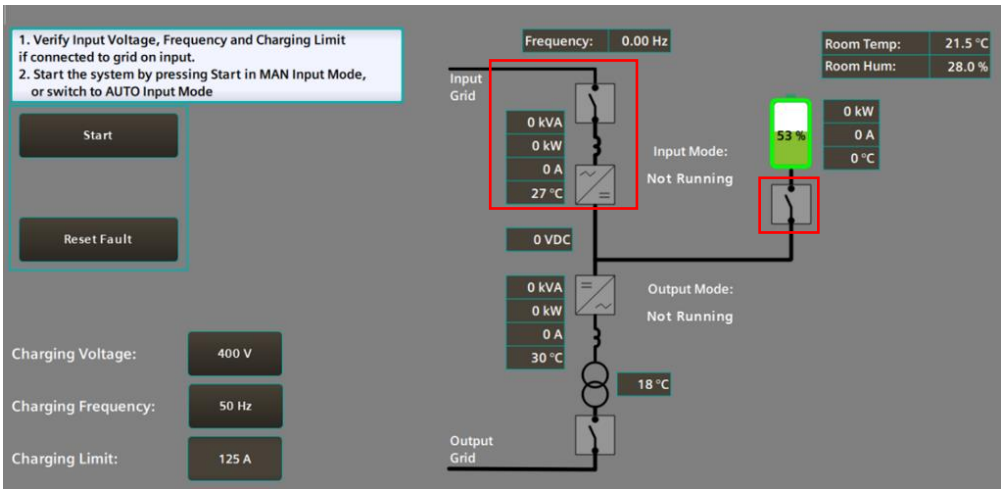
1. Verify that the output module is shut down. If yes, then move on to step 2. If the VFD or output module is active, go to section 4.3.6 “Output Module (DEIF) / Stop Sequence (Shutdown)” and follow the steps before returning.



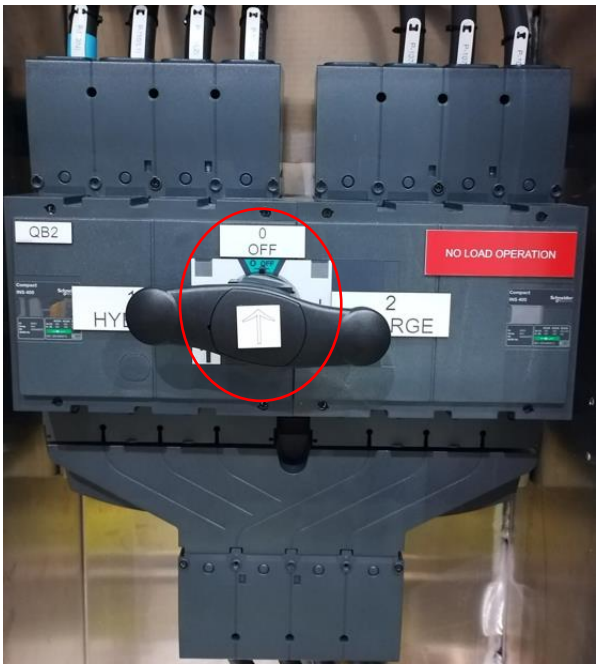
2. Press the “stop” button on the left side on the HMI Display (PSW).



3. Wait until the shutdown process has been completed and the green operating display status is stopped.

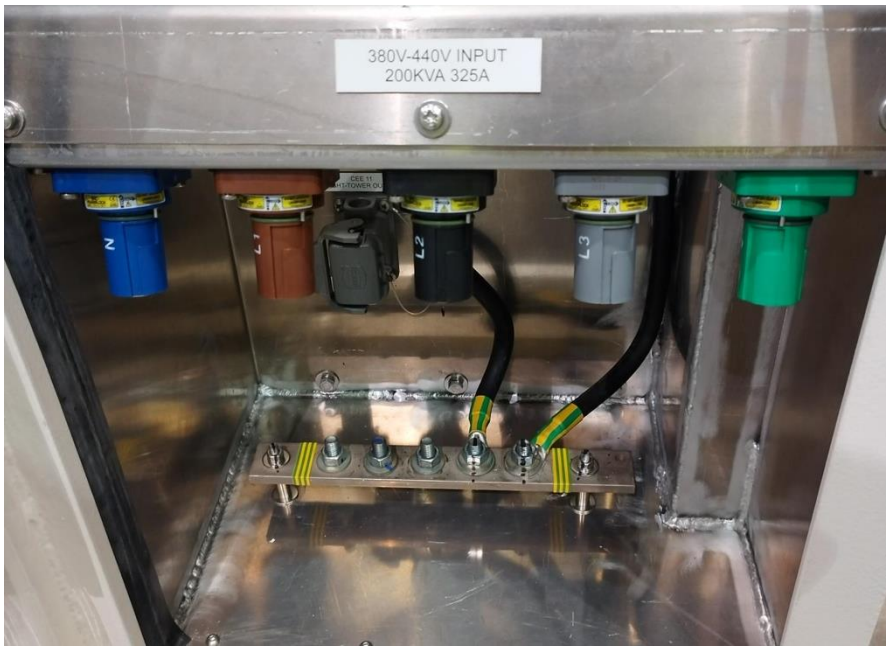


4. If the system was operated in “ongrid mode” and Powerlock selector switch (QB2) or CEE Switch (QB1) was activated, switch them back to the zero position.





5. Disconnect the cables from the input and output sockets and bar.





	INFORMATION If the grid power supply is connected, the unit will change operational mode to “On-Grid”. In the transition between “Off-Grid” and “On-Grid” mode the cooling and heating system will be restarted.

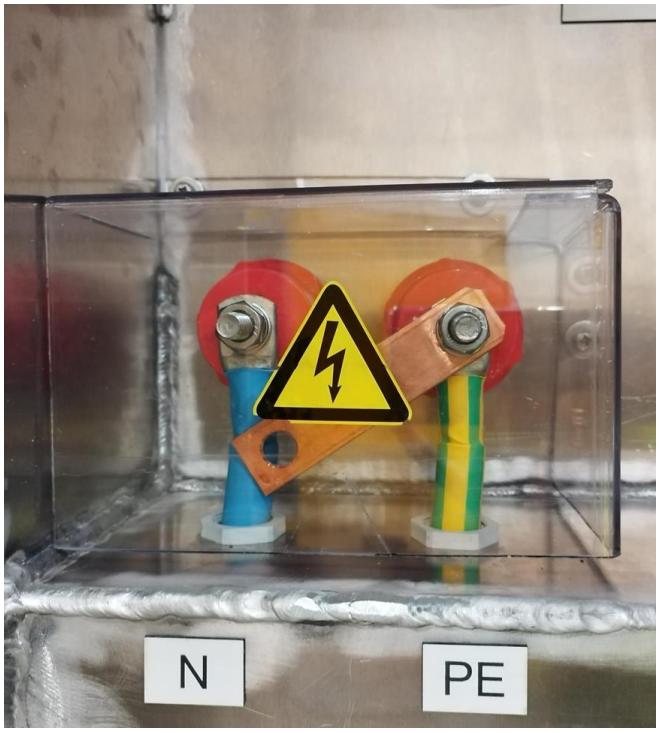
4.3 Output Operation (DEIF HMI + Control Panel)

4.3.1 Output Module (DEIF) / "Island" Mode

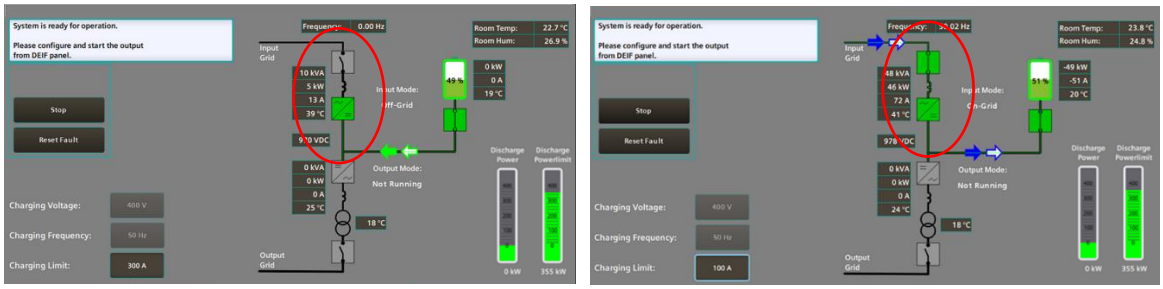
The DEIF ASC Battery unit controls the output module. By default, the Output Power module is in droop mode. The DEIF unit controls the Output Power Module frequency and voltage, in addition to the output AC breaker.

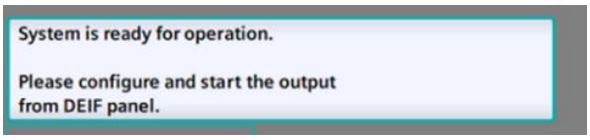
1. Connect cables to the AC Out section. For IT network the N-PE connection must be removed. For TN-network the N-PE connection must be made. 400kVA is the maximum rating at 380-690V 50-60Hz and 200kVA for 208-240V 50-60Hz (<45 min).





2. Make sure that the input module is running in “OFF-Grid” or “ON-Grid” mode.



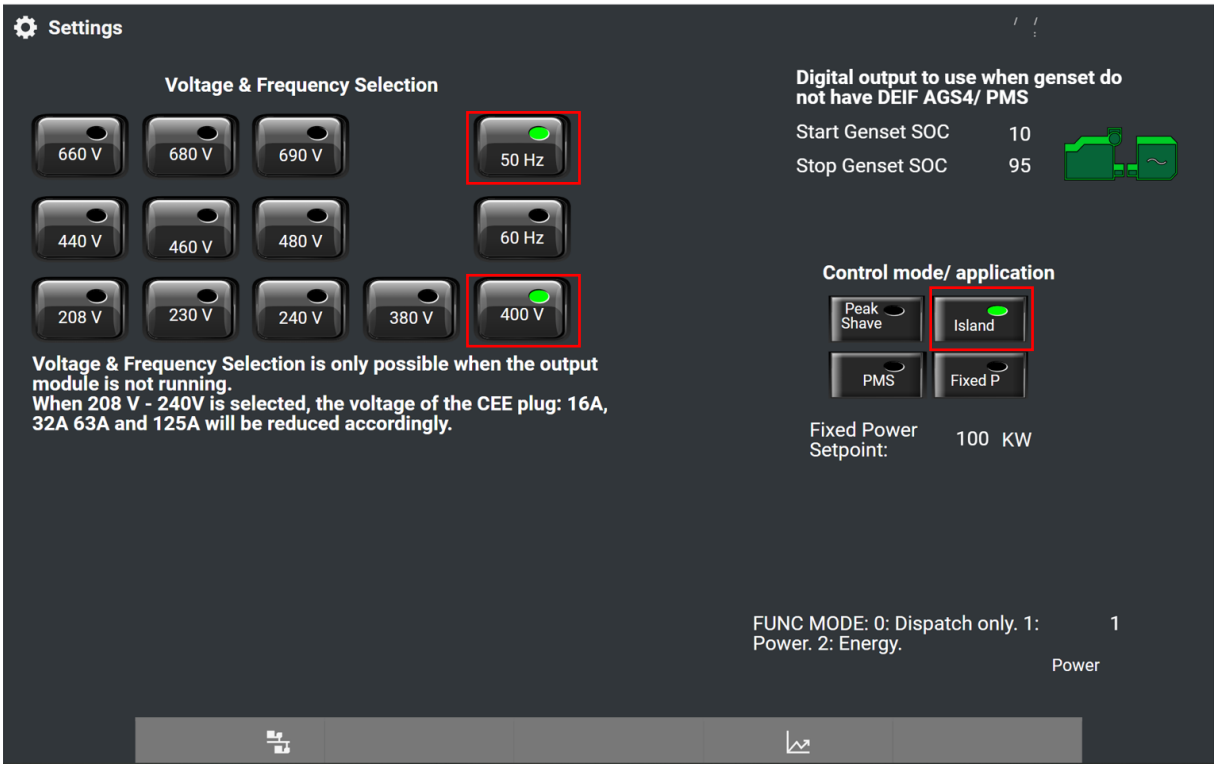


If not, you must check and see the status in section 4.2 “Input Module (PSW).

3. Select the correct voltage and frequency on the DEIF HMI Panel. To do this, press the area with the selection buttons and a new screen will open. This must be done before starting the output module.



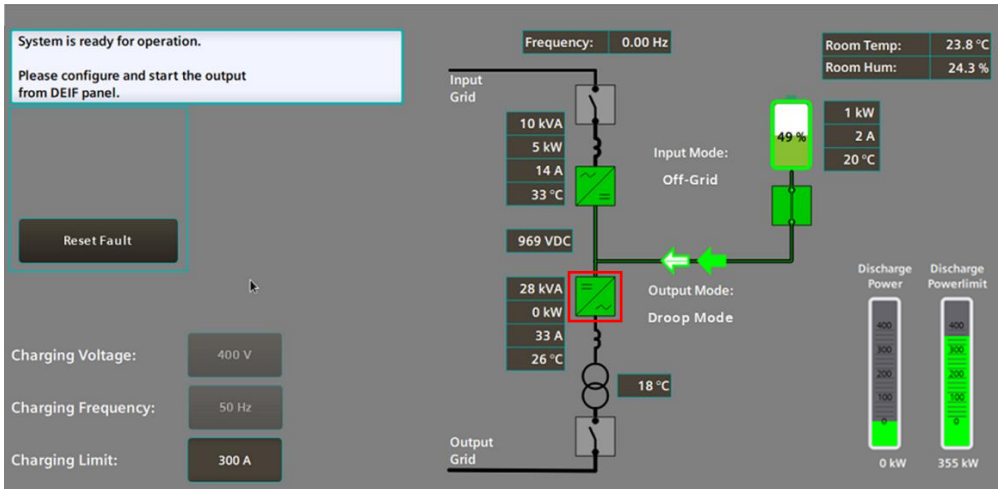
4. Press the "island" button for control mode.
Next, on the left-hand side of the screen, select the buttons for the required voltage and frequency. 400V / 50Hz was selected for this example.



5. Press “Start” on the DEIF ASC control panel.



The output module is now started and displayed in green as the operating state on the HMI display (PSW).



6. Verify green light on “Run” and “voltage/frequency” (green lights).
Before the load output is activated, you can check here again whether the selected voltage and frequency are correct.



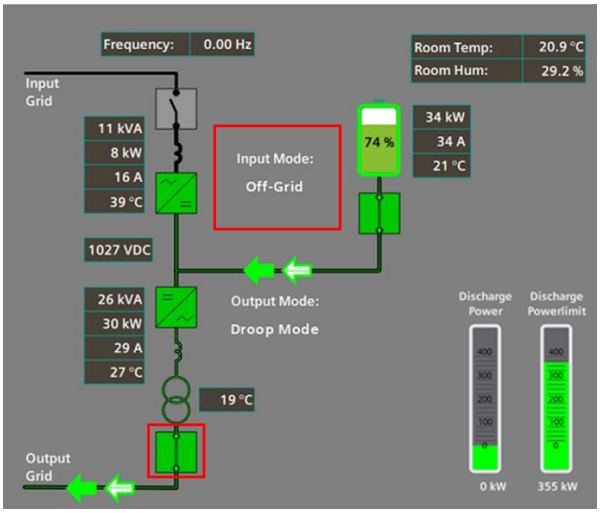
7. Press the close/open button to close the main AC breaker. Depending on the voltage selection these are the following breakers: Q5 (208-400V), Q6 (440-480V) and Q7 (660-690V).

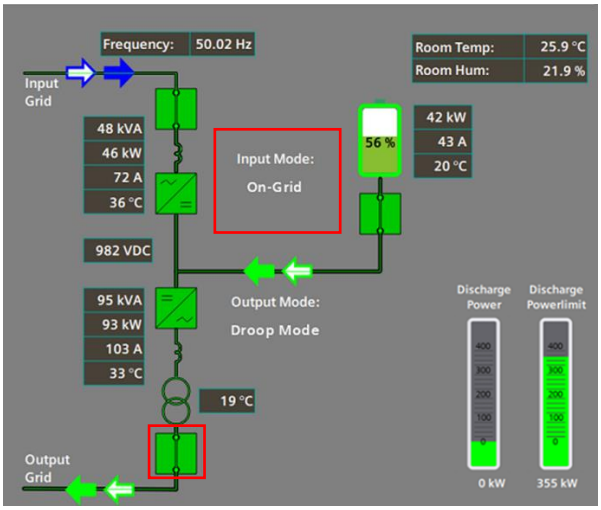


8. Verify that the bus voltage is correct (green light).



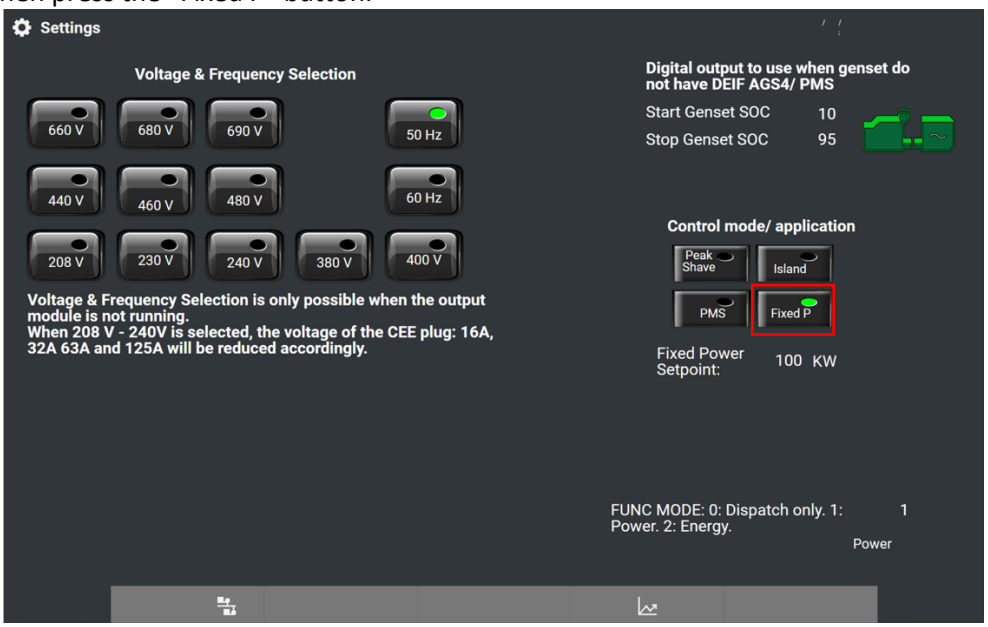
The main AC breaker (Q5, Q6, Q7) now turns green on the HMI display (PSW) and the direction of energy flow is shown in the direction of the load output.



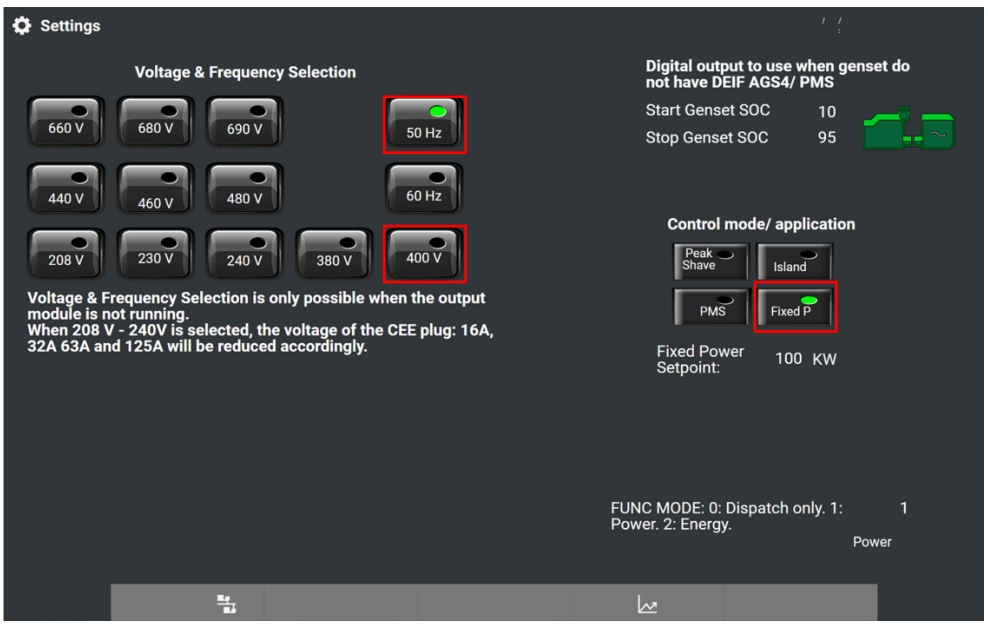


4.3.2 Output Module (DEIF) / "Fix Power" Mode

1. Start in the same way as in section 4.3.1. The points 1-3 are the same.
   Then press the "Fixed P" button.



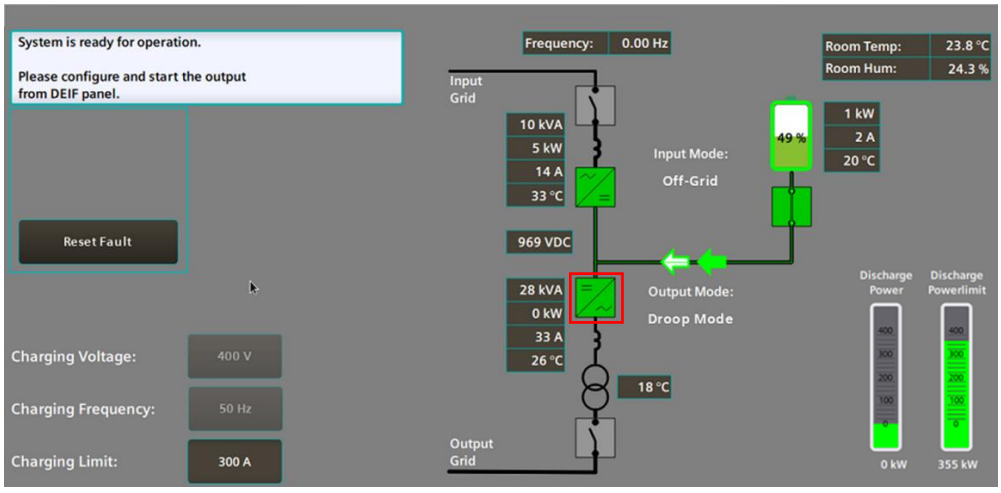
2. Select the fixed power (kW) that is to be provided on the output module.
   And press your selection for voltage & frequency.



3. Press “Start” on the DEIF ASC control panel.



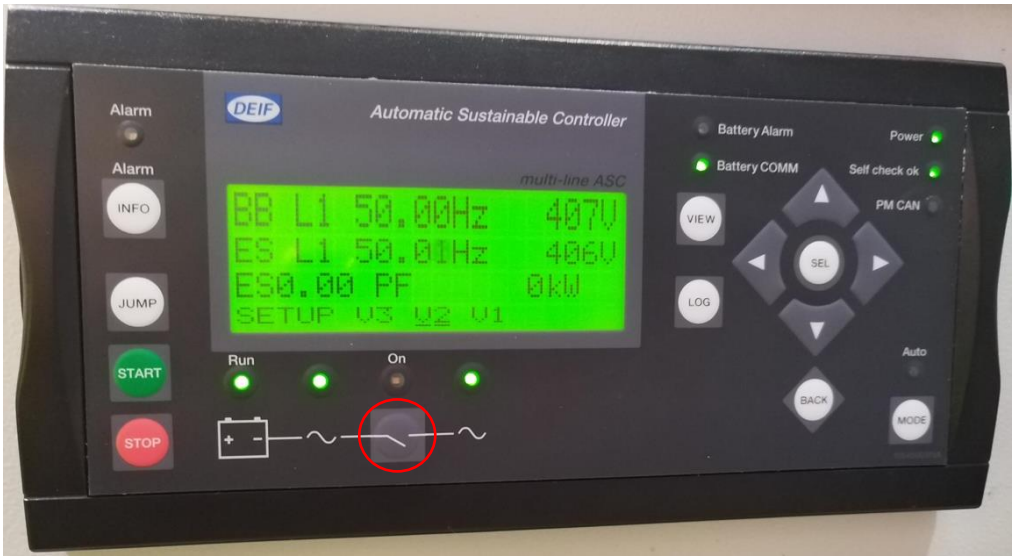
The output module is now started and displayed in green as the operating state on the HMI display (PSW).



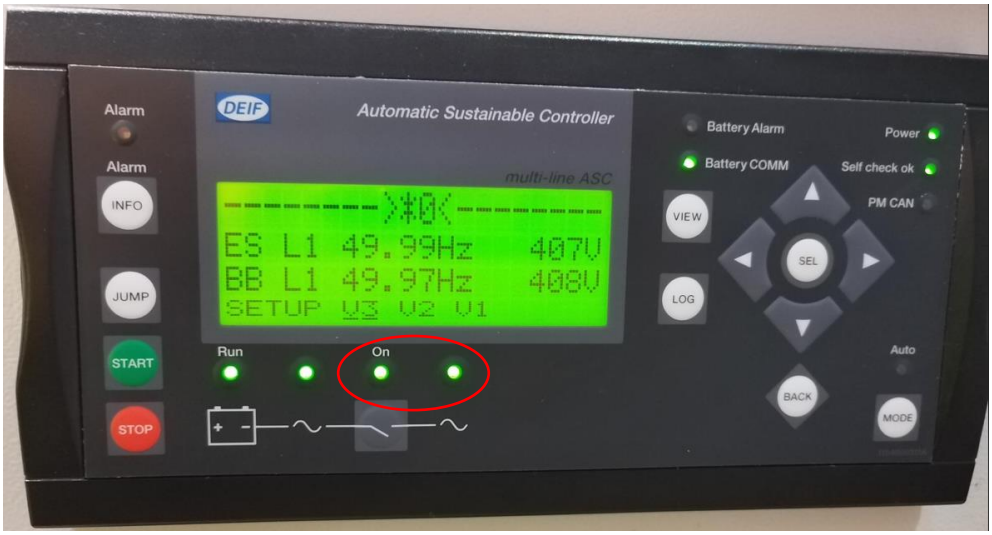
4. Verify green light on “run”, “voltage/frequency” and the grid (green lights).
Before the load output is activated, you can check whether the selected voltage and frequency are correct.



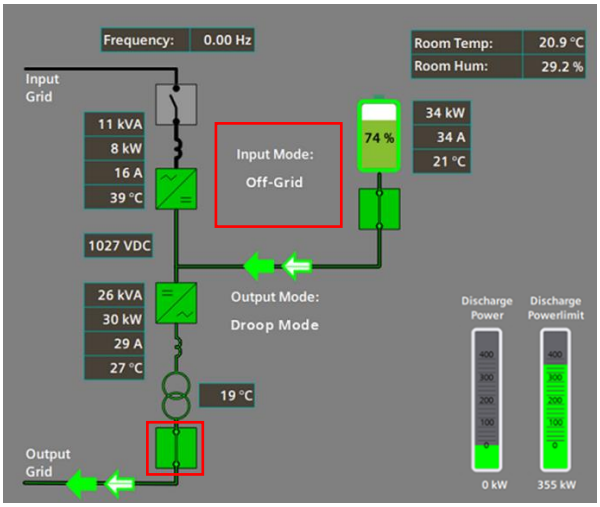
5. Press the close/open button to close the main AC breaker.

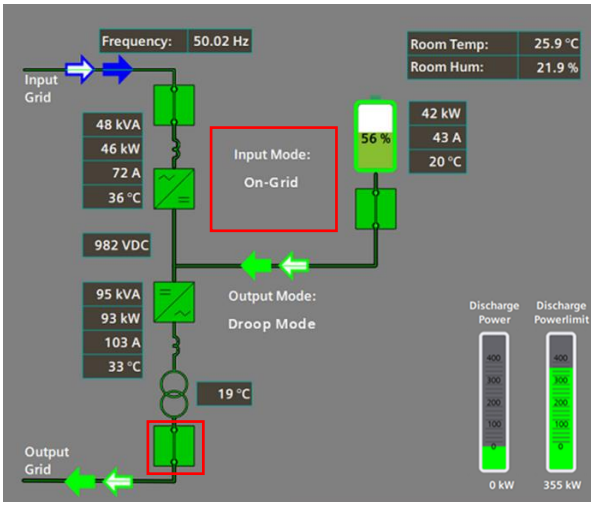


6. After synchronizing, the main AC breaker (Q5, Q6, Q7) is closed, and the LED is permanently on
(green light). Verify that the bus voltage is correct.



The main AC breaker (Q5, Q6, Q7) now turns green on the HMI display (PSW) and the direction of energy flow is shown in the direction of the load output.





4.3.3 Output Module (DEIF) / "Peak Shave" Mode

"Peak shaving" refers to leveling of peak electricity used by power consumers in the industrial and commercial sectors. Peak shaving reduces the amount of energy purchased from the utility companies during peak energy demand hours. For using this mode follow the previous steps from section 4.2.1 and read the manual from DEIF "ASC-4 Battery Designers Handbook" and "ASC-4 Commissioning Guidelines".



4.3.4 Output Module (DEIF) / "PMS" Mode

In the "PMS" mode, the individual controllers are in a small smart virtual grid for controlling the power sources (grid connection, gensets, batteries, …) The controllers constantly communicate over a network, exchanging information about the current load, their own power production capabilities, and the size of the current load on the power source they are controlling. Based on this information, the PMS calculates which power sources to use and how many power sources need to be active, and sends out commands as needed, for example to open and close breakers, crank gensets, …

To use this mode, follow the previous steps from section 4.2.1 and read the manual from DEIF "ASC-4 Battery Designers Handbook" and "ASC-4 Commissioning Guidelines".



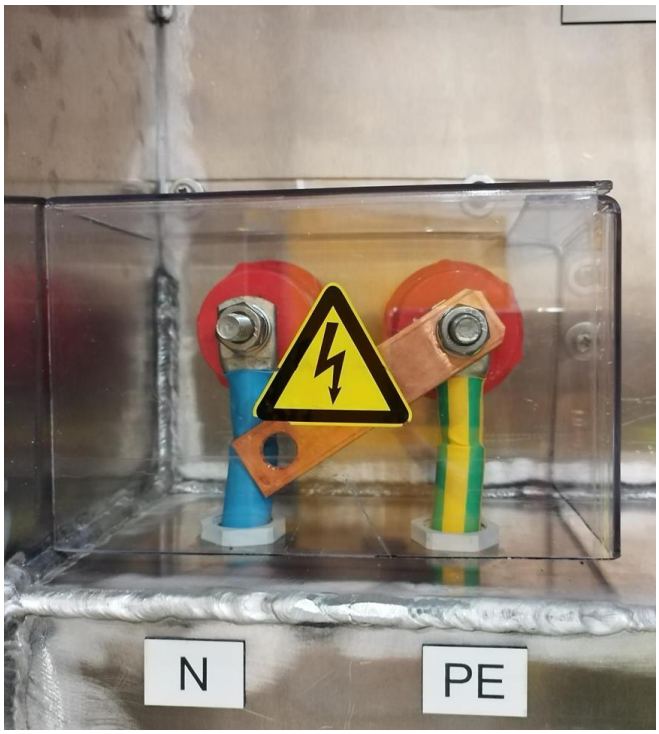
4.3.5 Output Module (DEIF) / On-grid (Parallel)

The unit can be operated in parallel with a grid or generator(s). This can be done in two ways

- Connecting the grid or generator(s) to the AC input (Island). Please see previous chapters describing the setup (SELECTOR SWITCH QB2 NEEDS TO BE IN HYBRID POS.)
- Connecting the grid or generator to the AC output (Parallel). Please follow the instructions below

1. Connect the power locks to the AC Out section. For IT network the N-PE connection must be removed. For TN-network the N-PE connection must be made.





⚠	WARNING To be able to do parallel operation on the AC output, the grid or generator(s) must have a dedicated DEIF controller and the PMS setup must be performed by qualified personnel.



ASC-4 Battery
Automatic Sustainable Controller





ASC-4
Automatic Sustainable Controller



4.3.6 Output Module (DEIF) / Stop Sequence (Shutdown)

1. Before opening the main AC breaker (Q5, Q6, Q7), de-load the unit, stop the load consumers and press the close/open button.

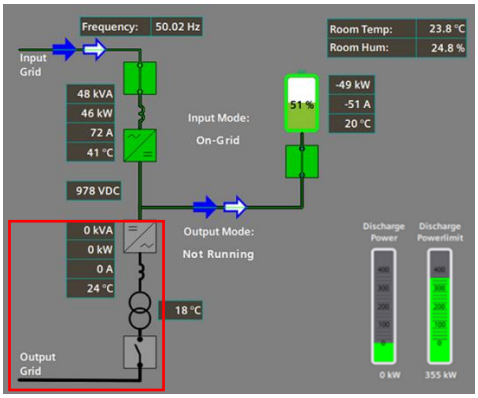


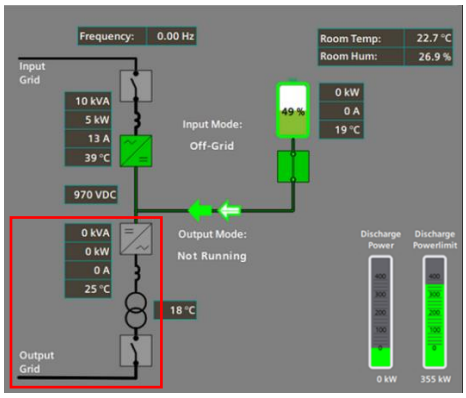
2. Press “Stop” on the DEIF ASC control panel to stop the output power module.



3. Verify that the green lights switch off (“run” and “voltage/frequency”) on the DEIF Control Panel and the status on the HMI Display (PSW).

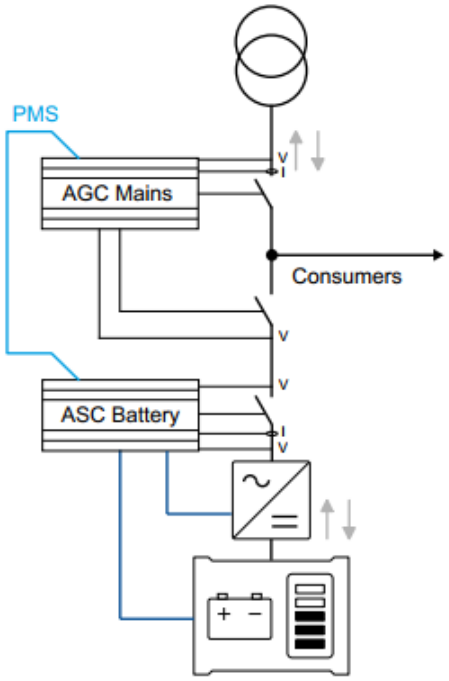




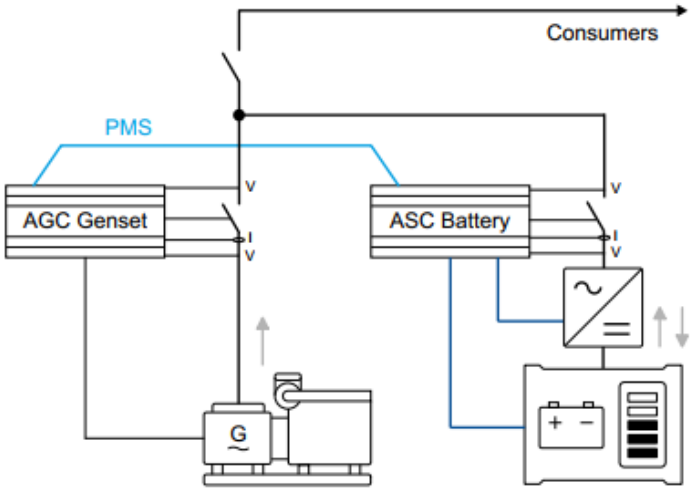


4.3.7 Examples of system configurations (DEIF)

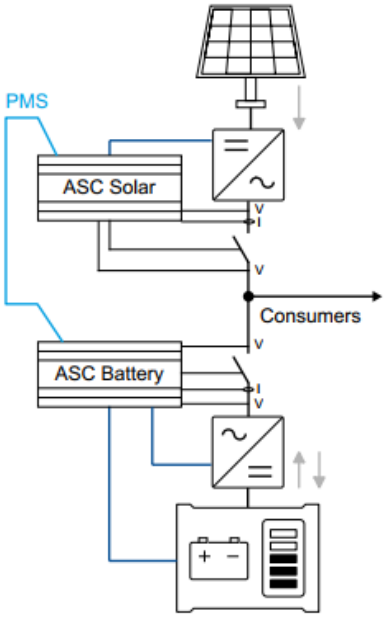
Grid + Battery



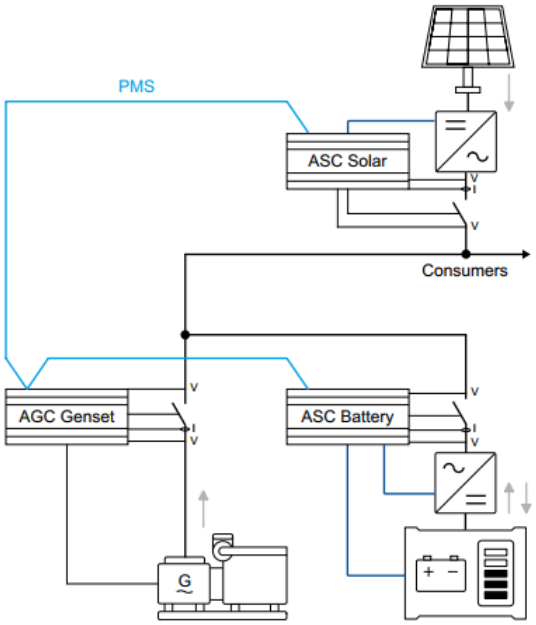
Generator + Battery



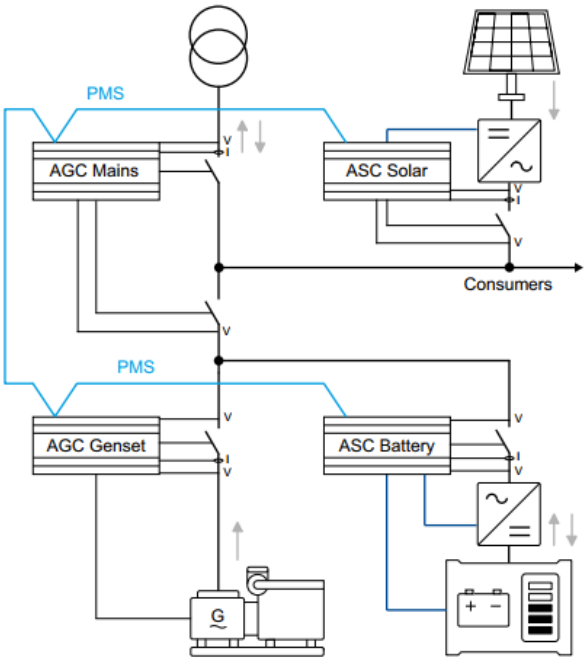
Solar + Battery



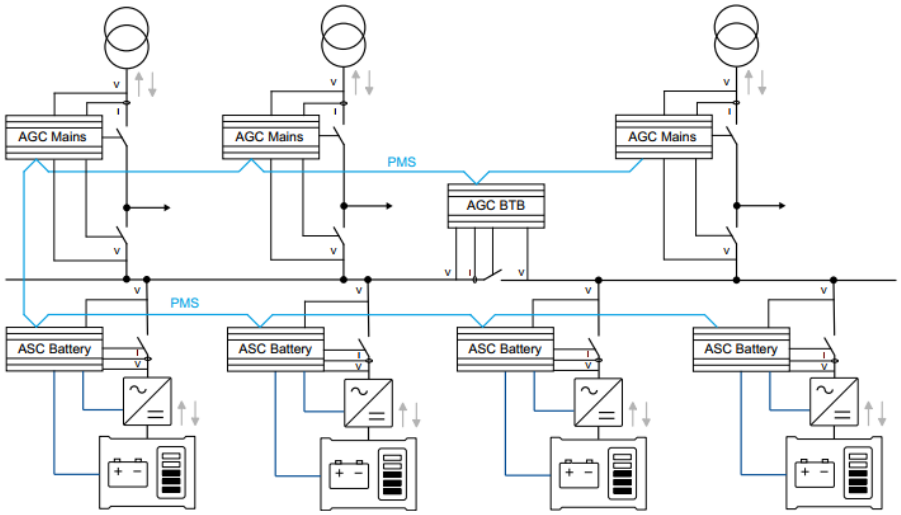
Generator + Solar + Battery



Generator + Solar + Battery + Grid



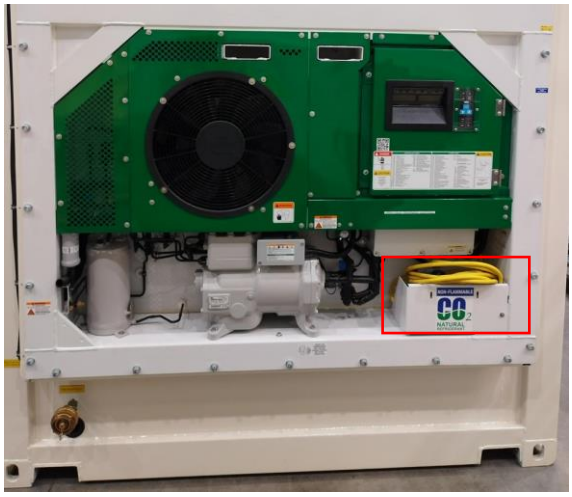
Bigger Hybrid Systems



5 Handling & Transport

5.1 Shutdown and preparing for transport

1. Following the steps from section 4.3.6 / Output module – stop sequence.
2. Following the steps from section 4.2.4 / Input module – stop sequence.
3. Switch “off” all main breakers.
4. Disconnect the output loads.
5. Make sure that there is no dangerous electrical energy in the supply connection from grid (input) and disconnect.
6. Place the cable for the reefer unit in the storage basket in the back of the container. Fasten it to make sure that it cannot fall out during transport.



7. Check that there are no loose objects inside the container. Close all doors (inside and outside).
8. Prepare lightning protection rod for transport by folding it in and securing it. See section 3.1.5 “Lightning Protection Rod”
9. Detach signal tower from the container, place it within the same basket as the reefer cable and secure it so that it won’t fall out during transport.

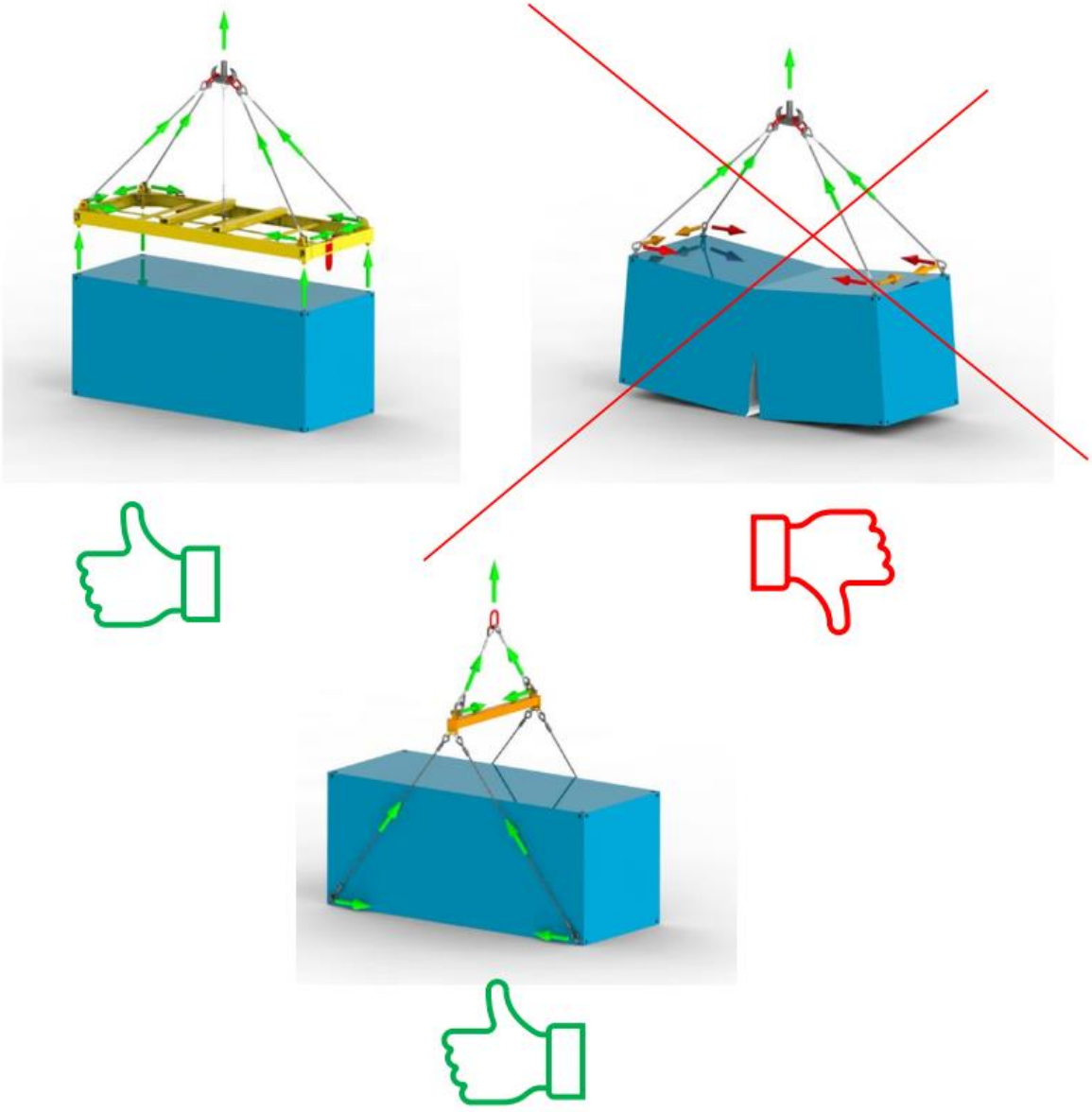
5.2 Transport and Preparation before Startup

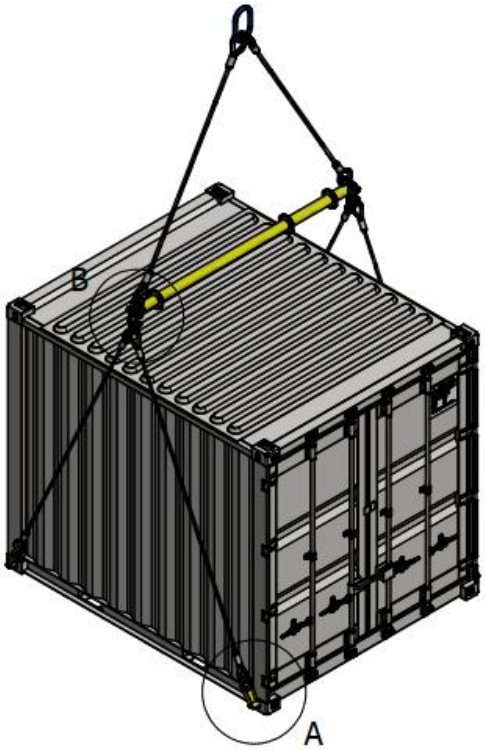
The BQ-S design is based on experience on transportable units in other segments, battery manufacturers own mobile units with the same batteries and customer input during sales, engineering and building phase. The design is sufficient for periodic transport on gravel roads and roads with asphalt. We as the manufacturer recommend that you check after each transport and ensure that no mechanical connections have become loose or that there are no loose parts inside the container.

5.3 Handling Container

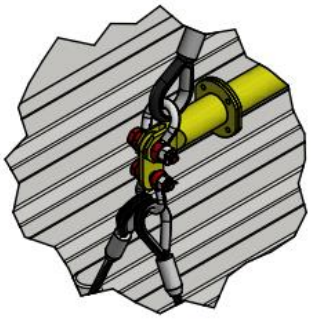
5.3.1 Crane Lifting

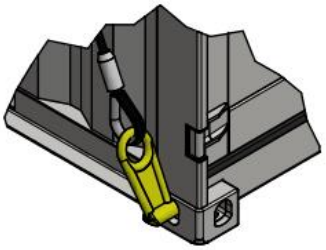
- DO NOT lift using the containers top corners (Chains, wires, etc)
- Lift from bottom with truck.
- Lift from bottom with MK-5 lifting gear.
- Lift from top with approved frame
- Lifting weight 10 ton

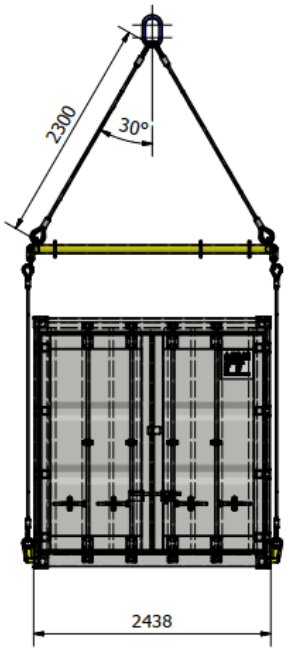


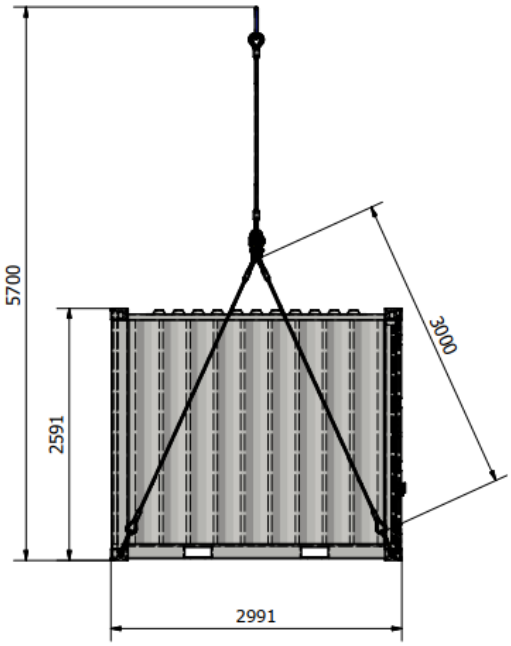


Typical rigging plan for the lifting of ISO containers
Use of 4 off ISO-corner lifting lugs at ends.
Use of 1 modular spreader beam MODULIFT
Use of wire rope slings over/under spreader
Capacity of slings and spreader are suited to customer requirements.



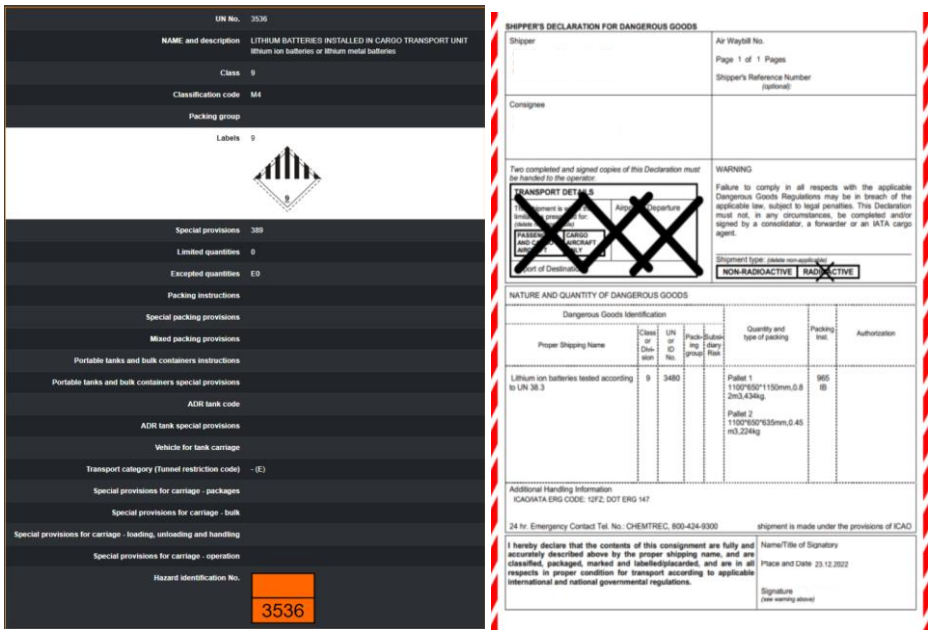






5.3.2 ADR Transport Document

ADR stands for Accord Dangerous Routier (European regulations concerning the international transport of dangerous goods by road.
Check the signs, stickers and the documents for your transport which are needed.



UN No.	3536
NAME and description	LITHIUM BATTERIES INSTALLED IN CARGO TRANSPORT UNIT lithium ion batteries or lithium metal batteries
Class	9
Classification code	M4
Packing group	
Labels	9
Special provisions	389
Limited quantities	0
Excepted quantities	E0
Packing instructions	
Special packing provisions	
Mixed packing provisions	
Portable tanks and bulk containers instructions	
Portable tanks and bulk containers special provisions	
ADR tank code	
ADR tank special provisions	
Vehicle for tank carriage	
Transport category (Tunnel restriction code)	- (E)
Special provisions for carriage - packages	
Special provisions for carriage - bulk	
Special provisions for carriage - loading, unloading and handling	
Special provisions for carriage - operation	
Hazard identification No.	3536

SHIPPER'S DECLARATION FOR DANGEROUS GOODS

Shipper

Air Waybill No.

Page 1 of 1 Pages

Shipper's Reference Number (optional)

Consignee

Two completed and signed copies of this Declaration must be handed to the operator

WARNING

Failure to comply in all respects with the applicable Dangerous Goods Regulations may be in breach of the applicable law, subject to legal penalties. This Declaration must not, in any circumstances, be completed and/or signed by a consolidator, a forwarder or an IATA cargo agent.

TRANSPORT DETAILS

CARGO AIRCRAFT ONLY

Shipment type: (delete non-applicable)

NON-RADIOACTIVE RADIOACTIVE

NATURE AND QUANTITY OF DANGEROUS GOODS

Dangerous Goods Identification: Proper Shipping Name	Class or Division	UN or ID No.	Packing group	Subsidiary Risk	Quantity and type of packing	Packing Inst.	Authorization
Lithium ion batteries tested according to UN 38.3	9	3480			Pallet 1 1100*650*1150mm,0.82m3,434kg. Pallet 2 1100*650*635mm,0.45m3,224kg	965 IB	

Additional Handling Information
ICAO/IATA ERG CODE: 12FZ; DOT ERG 147

24 hr. Emergency Contact Tel. No.: CHEMTREC, 800-424-9300 shipment is made under the provisions of ICAO

I hereby declare that the contents of this consignment are fully and accurately described above by the proper shipping name, and are classified, packaged, marked and labelled/placarded, and are in all respects in proper condition for transport according to applicable international and national governmental regulations.

Name/Title of Signatory

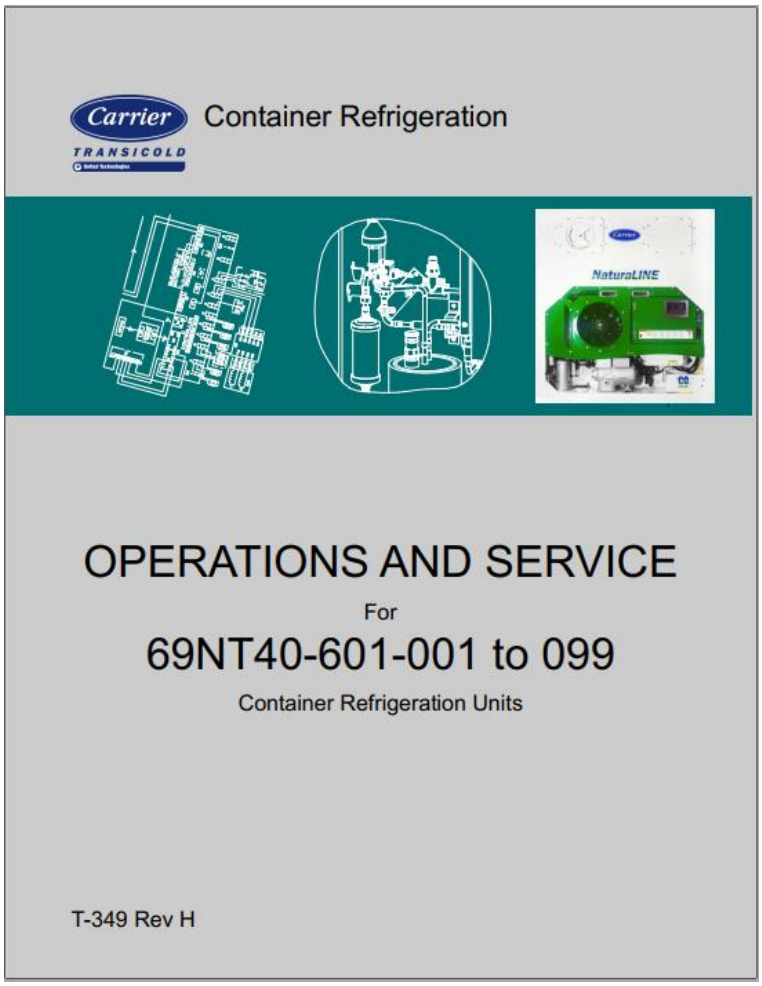
Place and Date 23.12.2022

Signature (see warning above)



6 Cooling & Heating

	INFORMATION The unit is cooled by CARRIER Container Refrigeration. Please see separate manual for this unit



7 Fault Handling

7.1 General troubleshooting

Fault	What to do…
Control Voltage Low Indicator (Red)	✓ Connect standby supply to charge
Low state of charge (SoC) warning from batteries	✓ Discharge is higher than charge power ✓ Reduce discharge power or increase charge power ✓ Connect charge supply and start input module in “on-grid” mode ✓ Check DEIF functionality if the unit is in parallel operation with the grid
Input or Output module Tripped	✓ Check alarm text ✓ Reset ✓ Re-start ✓ If issue persists, contact supplier
Battery partly disconnected or tripped	✓ Check alarm text ✓ Reset ✓ Shutdown ✓ Re-start ✓ If issue persists, contact supplier
Smoke or fire from the batteries or unit	✓ Contact local fire department ✓ Follow emergency procedures ✓ Contact supplier
Door Open alarm – Automatic shutdown	✓ Close the cargo doors ✓ Check the door switch on the top of the cabinet (right cargo door) ✓ Contact supplier
High temperature or humidity in container alarm	✓ Check the cooling and heating unit is on ✓ Check that the setpoint is set to 18 °C ✓ If issue persists, contact supplier
Low temperature alarm in container	✓ Check the cooling and heating unit is on ✓ Check that the setpoint is set to 18 °C ✓ If issue persists, contact supplier

High temperature in drive alarm	✓ Check that the cooling and heating unit is on ✓ Check that the setpoint is set to 18 °C ✓ Reduce load ✓ Check air fans and air flow and filters in drive ✓ If issue persists, contact supplier
High temperature in battery	✓ Check that the cooling and heating unit is on ✓ Check that the setpoint is set to 18 °C ✓ Reduce load ✓ Check air fans and air flow in battery modules ✓ If issue persists, contact supplier

7.2 HMI Display (PSW) PLC Controller Siemens

On the HMI Display for the input module (PSW) there is a general monitor and history for alarms and warnings (section 3.2.1 – status bar [#8] & section 3.2.3 – alarm list).

ID	Alarm Text	Action to take
1	Unable to communicate with BMS or BMS is unable to communicate with battery strings.	✓ Check that all the four RBMS's on the top of each battery rack are on. RBMS is on, this should be indicated by a green light on the round green on/off button ✓ Check that the CAN-bus communication cable from the black Kore Power unit inside the control panel is connected to the P3 socket on the RBMS in battery rack 1 is connected ✓ Check that the CAN-bus cable from the P4- socket on the RBMS in rack 1 is connected to the P3 socket on the RBMS in rack 2 ✓ Check that the CAN-bus cable from the P4 - socket on the RBMS in rack 2 is connected to the P3 socket on the RBMS in rack 3 ✓ Check that the CAN-bus cable from the P4 - socket on the RBMS in rack 3 is connected to the P3 socket on the RBMS in rack 4 ✓ Check that a CAN-bus resistor is connected to the P4 socket on the RBMS in rack 4¨ ✓ Reset ✓ If issue persists, contact supplier
2	BMS: Battery has exceeded maximum operating temperature	Alarm level ✓ Check that the cooling and heating unit is on ✓ Check that the setpoint is set to 18 °C ✓ Reduce load ✓ Check air fans and air flow in battery modules ✓ If issue persists, contact supplier

3	BMS: Battery is approaching maximum operating temperature.	Warning level ✓ Check that the cooling and heating unit is on ✓ Check air fans and air flow in battery modules ✓ Check that the setpoint is set to 18 °C ✓ Reduce load ✓ If issue persists, contact supplier
4	BMS: Battery has exceeded minimum operating temperature	Alarm level ✓ Check that the cooling and heating unit is on ✓ Check that the setpoint is set to 18 °C ✓ If issue persists, contact supplier
5	BMS: Battery is approaching minimum operating temperature.	Warning level ✓ Check that the cooling and heating unit is on ✓ Check that the setpoint is set to 18 °C ✓ If issue persists, contact supplier
6	BMS: Battery maximum charge current has been exceeded.	Alarm level ✓ Reduce charging though the output or the input module ✓ If issue persists, contact supplier
7	BMS: Approaching battery maximum charge current.	Warning level ✓ Reduce charging though the output or the input module ✓ If issue persists, contact supplier
8	BMS: Battery maximum discharge current has been exceeded.	Alarm level ✓ Reduce discharging load though the output module or increase charging though input module ✓ If issue persists, contact supplier

9	BMS: Approaching battery maximum discharge current.	Warning level ✓ Reduce discharging load though the output module or increase charging though input module ✓ If issue persists, contact supplier
10	BMS: Battery voltage has exceeded the maximum limit.	Alarm level ✓ Disconnect charging supply and start discharging though output module ✓ If issue persists, contact supplier
11	BMS: Battery voltage is approaching its maximum limit.	Warning level ✓ A warning will appear when approaching 100% State of Charge. This warning should disappear after discharge has started. ✓ If warning persist, contact supplier
12	BMS: Battery voltage has exceeded the minimum limit.	Alarm level ✓ Reduce discharging load though the output module or increase charging though input module ✓ If issue persists, contact supplier
13	BMS: Battery voltage is approaching the minimum limit.	Warning level ✓ Reduce discharging load though the output module or increase charging though input module ✓ If issue persists, contact supplier
18	BMS: A voltage imbalance exists between the strings in the battery bank.	Warning level ✓ After high power charge/discharge an unbalance alarm can occur ✓ Leave the system on with low load and the system will automatically balance itself
19	BMS: A temperature imbalance exists between the strings in the battery bank.	Alarm level ✓ After high power charge/discharge an unbalance alarm can occur ✓ Leave the system on with low load and the system will automatically balance itself

20	BMS: A temperature imbalance is developing between the strings in the battery bank.	Warning level ✓ A warning appears if the system is under high load over time. Cool air from the AC is cooling the lower battery modules, while heat is rising to the upper battery modules. ✓ Warning -Δ15 °C Alarm -Δ20 °C Trip -Δ 25°C ✓ Letting the AC unit run while system is under low load should balance the temperature
21	BMS: A contactor failed to open or close as requested.	✓ Reset ✓ Re-start ✓ If the issue persists, try to power cycle the 4. RBMS's on top of each battery rack with the round green on/off button ✓ Reset ✓ Re-start ✓ If issue persists after re-start, contact supplier
22	BMS: One or more battery fans have failed.	✓ Contact supplier
23	BMS: Ground fault detected.	✓ Shut down system ✓ Disconnect the Bender fuses in the DC-cabinet, -FU01, -FU02, FU03, -FU04. ✓ Turn off the 4. RBMS's on top of each battery rack with the round green on/off button ✓ Wait 10. Minutes for the DC to discharge ✓ Check for visible mechanical damage on power cables from battery racks to the DC-interface ✓ If no mechanical damage is found and issue persists, contact supplier
24	BMS: One or more doors are open.	✓ Contact supplier
25	BMS: A current imbalance exists between the strings in the battery bank.	✓ After high power charge/discharge an unbalance alarm can occur ✓ Leave the system on with low load and the system will automatically balance itself ✓ If issue persists, contact supplier

26	BMS: A vendor specific alarm has occurred.	✓ Contact supplier
27	BMS: A vendor specific warning has occurred.	✓ Contact supplier
28	BMS: The battery bank has been configured incorrectly and will not operate.	✓ Contact supplier
29	BMS: The battery bank has been configured incorrectly and may not operate as expected.	✓ Contact supplier
30	BMS: Battery cell voltage is approaching maximum limit.	✓ A warning could appear when approaching 100% State of Charge. Warning should disappear after discharge has started. ✓ If warning persists, contact supplier
31	BMS: Battery cell voltage is approaching the minimum limit.	✓ Reduce discharging load though the output module or increase charging though input module ✓ If issue persists, contact supplier
32	BMS: A voltage imbalance exists between the cells in the battery string.	✓ After high power charge/discharge an unbalance alarm can occur ✓ Leave the system on with low load and the system will automatically balance itself
33	Warning in frequency converter input module.	✓ Check input module display on the frequency converter for warning text
34	Frequency converter input module communication missing.	✓ Check that the network cable between input module in frequency converter and network switch in control cabinet is connected. Do not open frequency converter door with system energized! ✓ Reset ✓ If issue persists, contact supplier

35	Fault in frequency converter output module.	✓ Check output module display on the frequency converter for fault text. Faults like: -IGBT temp -Short circuit Can be caused by peaks in load. -Line Sync Fault Can be caused if converter failed to sync to grid. ✓ Reset ✓ If issue persists, or any other alarms appear contact supplier
36	Warning in frequency converter output module.	✓ Check output module display on the frequency converter for warning text
37	Frequency converter output module communication missing	✓ Check that the network cable between output module in frequency converter and network switch in control cabinet is connected. Do not open frequency converter door with system energized! ✓ Reset ✓ If issue persists, contact supplier
38	Fault in ESU.	✓ Reset ✓ If issue persist, contact supplier
39	Warning in ESU	✓ Reset ✓ If issue persists, contact supplier
40	PS1 supplying fan string 1,2 and 3 is not operating.	✓ Check that power supply -PS1 inside the control cabinet has 230V supply. ✓ If 230V is not present, check alarm list for tripped fuses ✓ If issue persists, contact supplier
41	PS2 supplying string 4 is not operating.	✓ Check that power supply -PS2 inside the control cabinet has 230V supply. ✓ If 230V is not present, check alarm list for tripped fuses ✓ If issue persists, contact supplier
42	PS3 charging the UPS is not operating.	✓ If system is in storage, check that 230V storage supply is connected ✓ Check that power supply -PS3 inside the control cabinet has 230V supply. ✓ If 230V is not present, check alarm list for tripped fuses ✓ If issue persist, contact supplier

43	Fuse -F2 or -FCZ.1 for the 230V control voltage has tripped.	✓ Fuse -F2 and -FCZ.1 are located in the middle section of the frequency converter. Check if tripped. Make sure the system is not energized before opening the frequency converter door. ✓ If issue persists, contact supplier
44	The emergency stop relay has been activated.	✓ Check the emergency stop button on control panel ✓ Check the emergency stop button on CEE 125A contact ✓ Reset ✓ Re-start ✓ If issue persists, contact supplier
45	Earth fault has been detected on the output grid >300mA.	✓ Check that earth fault monitoring controller inside the control cabinet is on. Screen should show level of earth fault ✓ If earth fault monitoring controller is on, check isolation resistance on output grid ✓ If earth fault monitoring controller is not on, check fuses ✓ If issue persists, contact supplier
46	Resistance in DC cable insulation is low.	✓ Shutdown ✓ Check isolation resistance ✓ Reset ✓ Re-start ✓ If issue persists, contact supplier
47	High current input module frequency converter.	✓ Reduce charging load ✓ If issue persists, contact supplier
48	Battery management system communication fault.	✓ Check that the network cable from the network switch to the black Kore Power box in the control cabinet is connected ✓ Check that the black Kore Power box in the control cabinet has 24V power supply, and that the “RUN” light is green ✓ Reset ✓ If issue persists, contact supplier
49	High current output module frequency converter.	✓ Reduce load though output module ✓ If issue persists, contact supplier
50	The state of Charge is low.	✓ Warning appears on <10% SoC ✓ Reduce discharge load though output module or increase charging though input module. ✓ If issue persists, contact supplier
51	SoC has reached shutdown level.	✓ The system has reached <1% SoC and shut down. ✓ Connect charging supply to recharge the system

		✓ Reset when SoC > 2% ✓ If alarm persists, contact supplier
53	The DC-breaker "-DC.Q1" failed to close within the time limit.	✓ Check that the control voltage fuses in DC-cabinet are on ✓ Check that DC-breaker switch is set to AUTO ✓ Reset ✓ If issue persists, or any other alarms appear contact supplier
55	Trafo temperature is high.	
56	Earth fault has been detected on the output grid > 270mA.	✓ Check that earth fault monitoring controller inside the control cabinet is on. Screen should show level of earth fault ✓ If earth fault monitoring controller is on, check isolation resistance on output grid ✓ If earth fault monitoring controller is not on, check fuses ✓ If issue persists, contact supplier
57	Fault in drive input module.	✓ Check input module display on the frequency converter for fault text. Faults like: -IGBT temp -Short circuit Can be caused by peaks in load. -Line Sync Fault Can be caused if converter failed to sync to grid. ✓ Reset ✓ If issue persists, or any other alarms appear contact supplier
58	The temperature in the container is above the warning limit.	✓ Check that the cooling and heating unit is on ✓ Check that setpoint is set to 18 °C
59	The temperature in the container is below the low warning limit.	✓ Check that the cooling and heating unit is on ✓ Check that setpoint is set to 18 °C
65	DEIF communication missing.	✓ Check that network cable from network switch in AC cabinet to DIEF control unit is connected ✓ Check that 24V supply for DEIF control unit is energized ✓ If issue persists, contact supplier
73	Resistance in DC cable insulation is too low.	✓ Shutdown ✓ Check isolation resistance ✓ Reset ✓ Re-start

		✓ If issue persists, contact supplier
74	The storage temperature in the container has exceeded the max recommended value.	✓ Check that the cooling and heating unit is on ✓ Check that setpoint is set to 18 °C ✓ If the system just has been turned off, let the cooling system work for 1-2 hours and see if the alarm disappears ✓ If issue persists, contact supplier
75	The storage temperature in the container has exceeded the min recommended value.	✓ Check that the cooling and heating unit is on ✓ Check that setpoint is set to 18 °C ✓ If issue persists, contact supplier
76	The storage humidity in the container has exceeded the max recommended value.	✓ Check that the 230V storage supply is connected ✓ Check that the dehumidifier is connected to the dehumidifier 230V socket, on switches is on and that the unit is running ✓ If issue persists, contact supplier
77	The container door is open. The system will soon automatically shut down.	✓ Check that the proximity switch on the outside of the control cabinet above the HMI is working. This can be checked by placing a piece of metal in front of the switch ✓ If the proximity switch is working, adjust the switch to make it detect the closed container door ✓ If issue persists, contact supplier
82	Fuse -F1 or -F3 for the cooling system has tripped.	✓ Fuses -F1 and -F3 are located in the middle section of the frequency converter. Check if tripped. Make sure the system is not energized before opening the frequency converter door. ✓ If issue persists, contact supplier
83	Fire detected.	✓ Check if there is smoke inside the container ✓ If smoke is detected, contact local fire department ✓ Follow emergency procedures ✓ Contact supplier ✓ If smoke is not detected, check that the fire detector is correctly mounted in the socket
84	Fuse in the DC-interface has tripped.	✓ This alarm indicates that either a high-speed DC fuse or a control fuse in the DC cabinet has tripped ✓ Check if any of the control fuses has tripped

		✓ To check if any high-speed DC fuses have tripped, the system needs to be de-energized. Turn off the system, turn off the high speed and control fuses for Bender in DC cabinet, and turn off the four RBMS units on the top of the battery racks by pushing the round green on/off button. Wait for 10 minutes to let the system de-energize and measure for voltage before checking the fuses. ✓ The status of the high-speed DC fuses is monitored by limit switches mounted on each fuse. Check that all limit switches are mounted correctly. ✓ Measure if any high-speed DC fuses are tripped by measuring the resistance though each fuse. Resistance should be close to 0Ω ✓ If issue persists, contact supplier
	BQ system is off and the temperature in the container is >30°C for more than 20 minutes.	✓ The container storage temperature is too high, the fans from the battery strings are shut down. ✓ The temperature in the container will get higher due to the battery fans running and the battery fans will increase its speed due to higher temperature. ✓ As the container is not getting cooled, the temperature will just keep rising. ✓ To avoid this, the battery fans are disconnected if the BQ system is off and the temperature in the container is >30°C for more than 20 minutes. ✓ If the reset button is pushed, the 20min timer will be reset. Also, if the temperature falls below 30°C or the system is started the battery fans will be reconnected.

7.3 Power Management – DEIF ASC Battery Controller

	INFORMATION The unit contains DEIF ASC battery controller with power management possibilities Please see separate manual for this unit for fault finding



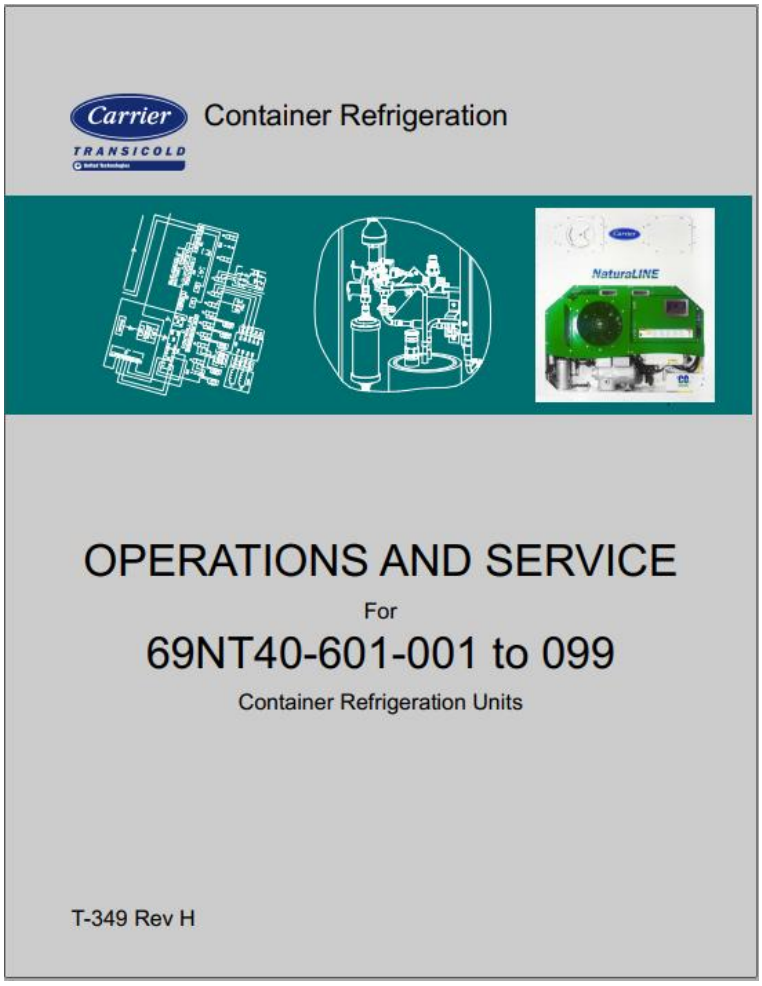
ASC-4

Automatic Sustainable Controller



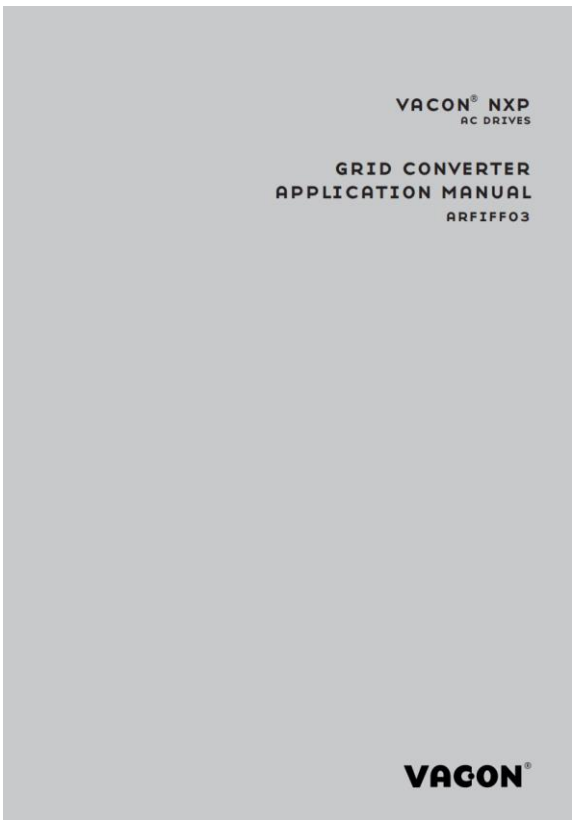
7.4 Cooling & Heating Unit

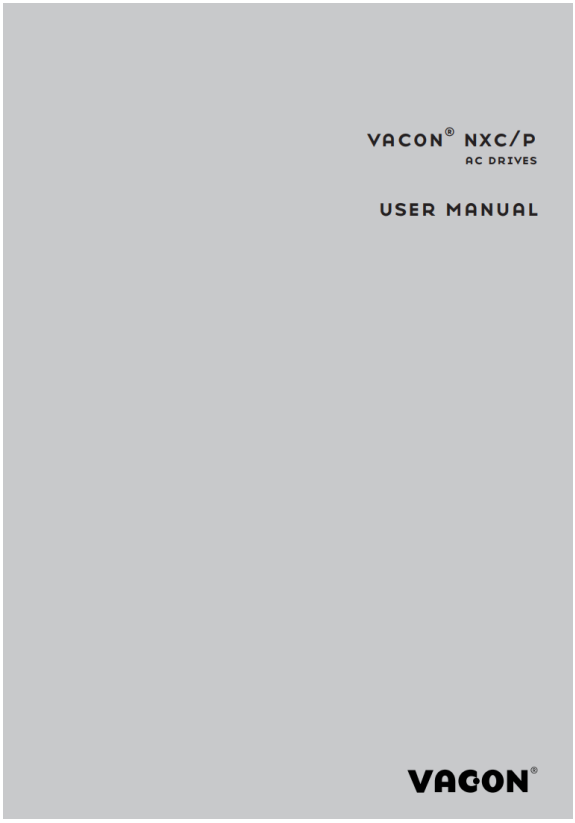
	INFORMATION The unit is cooled by Carrier Refrigeration. Please see separate manual for this unit for fault finding



7.5 Drive power modules

	INFORMATION The unit contains Danfoss Drive Please see separate manual for this unit for fault finding





8 Maintenance

Periodic maintenance is important to keep the system in good condition. The unit is equipped with a cooling & heating system and dehumidifier. This ensures an increased lifespan of expensive components and ensures problem-free operation.
Minimal inspection and maintenance are required to ensure trouble-free operation.

	Risk of personal injury. Only qualified personnel must carry out maintenance on the unit. Before starting maintenance, switch off the power supply to the unit. Use suitable locking equipment to isolate voltage. Follow the owner's locking procedure. The product is used in combination with a permanent energy source (battery). Even if the equipment is switched off, a dangerous electrical voltage may occur at the input and/or output terminals.

8.1 Plug and socket maintenance schedule

Plug and socket maintenance

No	Description	1 Month	6 Months	12 Months
1	Visual check	X		
2	Control of connections			X
3	Inspection of gaskets and O-rings		X	
4	Check function of all springs		X	
5	Check female contacts and clean if necessary.	X		
6	Check that the insulation is undamaged, and the correct torque is used.		X	
7	Check termination and cables in the cabinet.		X	
8	Inspect plug and plug housing for damage		X	
9	Check that the operating lever works satisfactorily.		X	

8.2 6-Months Maintenance

Technical Room	▪ Check the fastenings of all equipment. Tighten bolts as necessary. ▪ Check unit for water ingress. There should be no water in room ▪ Check covers, cabinet doors and nameplates
Electrical components	▪ Visual inspection of all components in cabinets. Protective cover and labeling must be in place. ▪ No loose objects inside the cabinets. ▪ Visual inspection of main cables. If there are signs of overheating in components, these must be replaced. ▪ Check that the protection on switches is correctly set. ▪ Open the housing on the transformer and check the connections. ▪ Check for corrosion and signs of overheating. Clean if necessary. ▪ Check labeling and warning signs and replace as necessary. ▪ Visual check and test fire detector with test-spray.
Cooling & Heating	▪ Check cooling and heating unit according to manual
Power Cables	▪ Inspect main cables for damage and replace as necessary. ▪ Check that there is correct cable protection in place and correct installation and fastening.
Drive	▪ Check I/O terminals ▪ Check tightness of mains connection ▪ Clean cooling tunnel ▪ Check operation of cooling fan, check for corrosion on terminals, busbars and other surfaces ▪ Check door and roof filters
Battery	▪ Inspect wires, terminals, and fasteners for corrosion ▪ Inspect covers properly operate and close to protect equipment ▪ Inspect all RMSC, Modules and Master BMS ▪ Confirm all fans are working as expected i.e., battery modules fans should operate at nearly

	the same speed at the same operating conditions ▪ Inspect alarm and fault logs to see if there are alarms or faults occurring regularly in the system indicating a need to replace a particular component ▪ Review torque on all electrical connections.

8.3 Documentation

After service, a service report should be issued with a review of check points. Repairs that have been carried out must be documented.

8.4 Contact

PSW Power & Automation is a Norwegian company that manufactures battery systems for land-based applications. The products are built in accordance with IIEC 62485 regulations. Over several years, the company has delivered reliable battery systems to consumers.

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